Canonical Descent and Formal Pro-Excision for Spectral Schemes

Abstract

We prove that topological Hochschild homology, regarded as a cyclotomic spectrum, and integral topological cyclic homology satisfy Čech descent for the canonical topology on connective spectral rings. The proof begins with canonical descent for finite symmetric powers of the spectral cotangent complex with bounded-below coefficients. It then passes through the complete circle-equivariant Hochschild–Kostant–Rosenberg filtration; linear connectivity of the filtration tails supplies continuity for finite-group Tate constructions, and naturality of the unfiltered cyclotomic Frobenius then upgrades the resulting comparison to cyclotomic descent.

As an application, we establish formal pro-excision for proper modifications of connective quasi-compact and quasi-separated spectral schemes. Let $Y \to X$ be proper and locally almost of finite presentation, and suppose that it is an equivalence over a quasi-compact open subscheme $U \subset X$. For both integral topological cyclic homology and nonconnective algebraic K-theory, the square obtained by completing X along $|X| \setminus U$ and Y along its inverse image is weakly cartesian in pro-spectra. The proof combines the finite case, comparison with animated formal completion over a classical base, derived pro-cdh descent, proper formal functions on derived loop spaces, and 1-connective pro-dévissage.

We also determine the scope of filtered descent. The complete Hodge filtration on Hodge-completed infinitesimal cohomology, the complete circle-equivariant Hochschild–Kostant–Rosenberg filtration on topological Hochschild homology and its finite-group Tate variants, and the p-adic filtration on derived p-complete topological cyclic homology satisfy canonical descent. By contrast, for every prime p, infinitely many positive associated graded pieces of the ambient Hahn–Raksit–Wilson even filtration fail to be sheaves for the canonical topology. On chromatically quasisyntomic rings these graded pieces are the Hahn–Raksit–Wilson syntomic complexes. The obstruction is chromatic: the p-complete sphere is a retract of $\mathrm{TC}(\mathbb{S})_p^\wedge$, whereas the multiplicative cyclotomic trace and Mitchell's theorem imply that $\mathrm{TC}(H\mathbb{Z})_p^\wedge$ is $K(2)$-acyclic.

1 Introduction

Let $W \to X$ be a canonical cover and let E be a contravariant spectrum-valued invariant. Canonical descent is the equivalence from $E(X)$ to the full derived Čech resolution

$$E(X) \longrightarrow \operatorname*{Tot}_{[q]\in\Delta} E\big(W^{\times_X (q+1)}\big). \tag{1.1}$$

Formal pro-excision asks for an additional compression of this resolution when W consists of a proper modification and a formal neighborhood of its closed complement. For a two-member cover $W = Y \amalg Z$, the degree-one term contains the derived self-intersections $Y \times_X Y$ and $Z \times_X Z$ as well as the two mixed intersections. The desired compression is the four-vertex map

$$E(X) \longrightarrow E(Y) \times_{E(Y\times_X Z)} E(Z) \tag{1.2}$$

and requires a separate argument; Proposition 6.5 gives a rational square-zero example where it is not an equivalence.

We first prove canonical descent for finite cotangent polynomials, for THH as a cyclotomic spectrum, and for TC. We then show that, for a proper modification, formal completion compresses the full Čech resolution to the four vertices of the completion square. This requires a uniform pro-argument, since a degreewise pro-equivalence need not remain a pro-equivalence after an infinite totalization.

1.1 Main results

Let $p : Y \to X$ be a proper morphism, locally almost of finite presentation, between connective quasi-compact and quasi-separated (qcqs) spectral schemes. Suppose that p is an equivalence over a quasi-compact open subscheme $U \subset X$, and let $Z = |X| \setminus U$. Formal completions below are computed from fixed compatible free spectral Koszul towers on the affine members of a finite Zariski–Mayer–Vietoris presentation. Ordinary spectra in the resulting formal squares are regarded as constant pro-spectra.

Theorem A. For $E \in \{K, \mathrm{TC}\}$, the square

$$\begin{array}{ccc} E(X) & \longrightarrow & E(X^\wedge_Z) \\ \big\downarrow & & \big\downarrow \\ E(Y) & \longrightarrow & E(Y^\wedge_{p^{-1}Z}) \end{array}$$

is weakly cartesian in Pro(Sp).

This is Theorem 11.1. For a formal tower $X_a \to X$ and $Y_a = Y \times_X X_a$, set

$$\Phi_E(p) = \Big\{ \mathrm{fib}\big(E(X) \longrightarrow E(Y) \times_{E(Y_a)} E(X_a)\big) \Big\}_a . \tag{1.3}$$

Proposition 7.3 identifies this pro-spectrum with the fiber of the canonical compression from the full Čech resolution to the four-vertex pullback. Theorem A is therefore equivalent to the vanishing of $\Phi_E(p)$.

The descent input is stated for a small canonical family $\{A \to B_i\}_{i \in I}$ of connective E_∞-rings. If $\alpha = (i_0, \ldots, i_q)$, write $B_\alpha = B_{i_0} \otimes_A \cdots \otimes_A B_{i_q}$.

Theorem B. For every small canonical family, the comparison

$$\mathrm{THH}(A) \longrightarrow \operatorname*{Tot}_q \prod_{\alpha \in I^{q+1}} \mathrm{THH}(B_\alpha)$$

is an equivalence in CycSp. Consequently

$$\mathrm{TC}(A) \longrightarrow \operatorname*{Tot}_q \prod_{\alpha \in I^{q+1}} \mathrm{TC}(B_\alpha)$$

is an equivalence. The complete circle-equivariant Hochschild–Kostant–Rosenberg (HKR) filtration and, for every finite $C_m \subset \mathbb{T}$, its termwise C_m-Tate filtration also satisfy filtered canonical descent.

The unfiltered assertion is Theorem 4.8; the filtered assertions are Theorem 4.3 and Corollary 4.6. The proof is built from Proposition 3.2, which establishes descent for finite cotangent polynomials

$$M \otimes_R \mathrm{Sym}^d_R(\mathbb{L}^{E_\infty}_{R/A} \otimes V)$$

with bounded-below coefficients and a finite-group Tate refinement. A finite transitivity filtration separates the base-change term from positive relative-cotangent terms; a second split Čech direction kills the latter. Complete filtered detection then reconstructs the HKR tower. Finally, the linear connectivity of its tails supplies precisely the continuity required to pass through finite-group Tate constructions.

The filtration on TC requires a separate discussion. The p-adic filtration on derived p-complete TC is complete, multiplicative, and satisfies canonical descent for every prime; see Theorem 5.1. The Hahn–Raksit–Wilson even or motivic filtration behaves differently.

Theorem C. Let

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(R) := \mathrm{gr}^i\,\mathrm{fil}^{\bullet}_{\mathrm{ev},p,\mathrm{TC}}(\mathrm{THH}(R)^{\wedge}_p)[-2i]$$

be the normalized associated graded piece of the ambient Hahn–Raksit–Wilson (HRW) even-TC filtration. For every prime p, there are infinitely many integers $i > 0$ for which

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(\mathbb{S}) \longrightarrow \mathbb{Z}_p^{\mathrm{ev}}(i)(H\mathbb{Z})$$

is not an equivalence, and $\mathbb{Z}_p^{\mathrm{ev}}(i)$ is not a sheaf for the canonical topology. Since $\mathbb{S}$ and $H\mathbb{Z}$ are chromatically quasisyntomic, the displayed values of the ambient even filtration are the corresponding HRW syntomic complexes.

This is Theorem 5.5. Each normalized ambient graded piece is uniformly coconnective. Consequently, if it were a sheaf for the canonical topology, it would be hypercomplete. Descent along the constant Postnikov hypercover of $\mathbb{S}$ by $H\mathbb{Z}$ would then identify its values on these two rings. The chromatic calculation below shows that this identification fails in infinitely many positive weights. The theorem does not assert failure in every positive weight, nor does it single out a particular one; the ambient negative pieces vanish identically.

The endpoint calculation is chromatic and uniform in the prime. The Bökstedt–Hsiang–Madsen splitting exhibits the p-complete sphere as a retract of $\mathrm{TC}(\mathbb{S})^{\wedge}_p$. On the other hand, multiplicativity of the cyclotomic trace makes $\mathrm{TC}(H\mathbb{Z})^{\wedge}_p$ a $K(\mathbb{Z})$-module, and Mitchell's theorem makes every such module $K(2)$-acyclic. Thus the relative TC spectrum is not $K(2)$-acyclic, including at $p = 2$. On the other hand, every normalized ambient graded piece of the even-TC filtration canonically lifts to an $H\mathbb{Z}$-module. Consequently each relative graded piece is $K(2)$-acyclic, so only finitely many nonzero relative weights would force the relative TC spectrum itself to be $K(2)$-acyclic, a contradiction.

1.2 Formal pro-excision

The proof of Theorem A proceeds through the defect (1.3). For a finite modification locally almost of finite presentation, supported almost-perfect completion identifies the defect module with its formal tower, and the connectivity form of the Land–Tamme pro-Milnor theorem gives formal pro-excision.

Over a classical base, relative animated reflection replaces Y by an animated scheme Y^{an} with the same classical truncation. The comparison $(Y^{\mathrm{an}})^{\mathrm{sp}} \to Y$ is a finite modification. The derived pro-cdh theorem of Kelly–Saito–Tamme compares the formal relative terms for X and Y^{an}, while the finite spectral theorem compares those for Y and $(Y^{\mathrm{an}})^{\mathrm{sp}}$; two-out-of-three proves the classical-base case. For a general spectral base, invariance of $\Phi_E(p)$ under the affine map $t_0X \to X$ reduces the theorem to the classical-base case.

The 1-connective invariance theorem has three inputs. Raskin's construction gives an integral complete filtration for relative TC of a nonsplit square-zero extension; its weight-m quotient is obtained by taking C_m-homotopy orbits of THH with coefficients, and the tails become linearly more connective with m. A marked cyclic-word model supplies the required C_m-action. Proper formal functions on derived loop spaces make each fixed-weight formal defect weakly zero, while its stagewise fibers admit a lower bound independent of the formal index. Filtered pro-dévissage then combines fixed-weight vanishing with the uniform tail estimate. Iterating the Lee–Levy factorization doubles connectivity and proves 1-connective invariance. The truncating fiber $K^{\mathrm{inv}} = \mathrm{fib}(K \to \mathrm{TC})$ transfers the conclusion from TC to nonconnective K-theory.

1.3 Relation to previous work

The cotangent input is the canonical descent theorem recorded by Antieau and attributed there to Antieau–Iwasa–Krause [1, Theorem 2.11 and Corollary 2.12]. We use its individual Hodge pieces as the associated graded of the equivariant HKR tower and then prove the continuity and cyclotomic assembly needed for THH and TC. The even and motivic filtrations are those of Hahn–Raksit–Wilson [10]. After composition with $R \mapsto \mathrm{THH}(R)^{\wedge}_{p}$, their even-TC filtration defines an ambient presheaf on all connective spectral rings. On the chromatically quasisyntomic subcategory they prove exhaustivity and identify its associated graded pieces as syntomic complexes. Their descent theorem applies along maps that are p-completely evenly faithfully flat (p-completely eff) in their terminology, whereas Theorem C shows that the presheaves of ambient graded pieces do not satisfy descent for the full canonical topology.

For formal pro-excision, the finite step uses the pro-Milnor theorem of Land–Tamme [13], while the classical-base step uses the derived pro-cdh theorem of Kelly–Saito–Tamme [12]. Iwasa has announced canonical descent for THH and TC, the resulting cdh descent of K-theory on spectral schemes, and spectral cdh pro-excision [11]. Thus Theorems A and B and Proposition 6.4 overlap with that announced package. The present treatment records the small-family and equivariant refinements explicitly, includes the filtered Tate-continuity and HRW obstruction results, and proves proper pro-excision through the formal-defect comparison, the spectral–animated completion comparison, proper formal functions on derived loop spaces, and the 1-connective invariance theorem of Sections 7–10.

1.4 Organization

Sections 2–5 develop canonical descent, cotangent and HKR filtrations, cyclotomic descent, and the HRW obstruction. Section 6 passes to spectral schemes and records the four-vertex counterexample. Section 7 defines the formal defect and proves the finite modification theorem. Sections 8– 10 establish the classical-base comparison, proper formal functions, and 1-connective invariance. Section 11 assembles these inputs into Theorem A.

2 Canonical families and linear descent

This section develops the linear input for cotangent descent. We record three consequences of canonicality: stability under base change and refinement, descent for bounded-below modules and finite-group constructions, and contraction when one member of a Čech family splits. All rings are connective E_∞-rings, and all tensor products are derived. If $\{A \to B_i\}_{i\in I}$ is a small family and $\alpha = (i_0, \ldots, i_q) \in I^{q+1}$, write

$$B_\alpha = B_{i_0} \otimes_A \cdots \otimes_A B_{i_q}.$$

We use the convention that a spectrum is n-connective when its homotopy groups vanish below degree n, and a map is n-connective when its fiber is n-connective.

2.1 Canonical families

Definition 2.1 (Canonical family)**.** The family $\{A \to B_i\}_{i\in I}$ is *canonical* if, for every connective A-algebra A', the augmentation

$$A' \longrightarrow \operatorname*{Tot}_{[q]\in\Delta} \prod_{\alpha\in I^{q+1}} (A' \otimes_A B_\alpha) \tag{2.1}$$

is an equivalence in $\mathrm{CAlg}^{\mathrm{cn}}$.

For an infinite family the product in (2.1) is taken after the individual tensor products have been formed. Thus the definition does not replace the family by the single map $A \to \prod_i B_i$. The word "small" refers only to the chosen Grothendieck universe.

Proposition 2.2 (Canonical-topology interpretation)**.** *Definition 2.1 is equivalent to saying that* $\{\operatorname{Spec} B_i \to \operatorname{Spec} A\}$ *is a covering family for the canonical Grothendieck topology on connective spectral affines.*

Proof. A family of affines is canonical precisely when it is universally effective epimorphic. After a base change $\operatorname{Spec} A' \to \operatorname{Spec} A$, a representable test object $\operatorname{Spec} C$ sees the map

$$\mathrm{Map}_{\mathrm{CAlg}}(C, A') \longrightarrow \operatorname*{Tot}_{q} \prod_{\alpha\in I^{q+1}} \mathrm{Map}_{\mathrm{CAlg}}(C, A' \otimes_A B_\alpha).$$

By the universal property of limits, this map is an equivalence for every connective C exactly when (2.1) is an equivalence. Yoneda proves both implications. □

The proposition concerns effective epimorphy after canonical sheafification. It does not assert pointwise effective epimorphy of the coproduct of the original representable presheaves.

We use two descent conventions. Unless the prefix "hyper" is written, canonical descent means ordinary family Čech descent as in Definition 2.1. A *canonical hypersheaf* is a presheaf satisfying descent for every hypercover in the canonical topology. Thus hypersheaf descent is an a priori stronger assertion on this site; the distinction will be essential in Theorem 5.5.

Lemma 2.3 (Family calculus)**.** *Canonical families are stable under connective base change and composition. If a canonical family* $\{A \to C_k\}_{k\in K}$ *refines* $\{A \to B_i\}_{i\in I}$ *through factorizations* $A \to B_{\varphi(k)} \to C_k$*, then* $\{A \to B_i\}$ *is canonical. In particular, if* $A \to B \to C$ *and* $A \to C$ *is canonical, then* $A \to B$ *is canonical. The same statements hold for connective commutative algebra objects in* $\mathrm{Fun}(BG, \mathrm{Sp})$ *for any group* G*. Moreover, a nonequivariant canonical family, equipped with trivial* G*-actions, is canonical in this equivariant category.*

Proof. After canonical sheafification, the map from the formal coproduct of the representables to the target is an effective epimorphism. Effective epimorphisms in an infinity-topos are stable under base change, coproducts, and composition. For a refinement there is a factorization

$$\coprod_{k\in K} y(\operatorname{Spec} C_k) \longrightarrow \coprod_{i\in I} y(\operatorname{Spec} B_i) \longrightarrow y(\operatorname{Spec} A).$$

The composite is an effective epimorphism, so the second arrow is one by right cancellation for the connected–truncated factorization system [16, Corollary 6.2.3.12].

Repeating the same argument in equivariant connective commutative algebras gives the corresponding formal properties. For the final assertion, let C be any connective equivariant A-algebra. After forgetting the action, Definition 2.1 gives

$$C \simeq \operatorname*{Tot}_{q} \prod_{\alpha} (C \otimes_A B_\alpha).$$

The forgetful functor creates limits and detects equivalences, so this is an equivariant Amitsur equivalence. The same argument applies after every equivariant connective base change. $\square$

Lemma 2.4 (Connectivity criterion)**.** *If $A \to B$ has 1-connective fiber, then the singleton family $\{A \to B\}$ is canonical.*

Proof. Let A' be a connective A-algebra, and put

$$B' = A' \otimes_A B, \qquad J' = \operatorname{fib}(A' \longrightarrow B').$$

Extension of scalars is exact, so $J' \simeq A' \otimes_A \operatorname{fib}(A \to B)$ is 1-connective. If

$$T'_n = \lim_{[q]\in\Delta_{\leq n}} (B')^{\otimes_{A'}(q+1)},$$

then the cubical formula for partial Amitsur totalizations gives a natural cofiber sequence

$$(J')^{\otimes_{A'}(n+1)} \longrightarrow A' \longrightarrow T'_n; \tag{2.2}$$

see [10, Proposition 2.3.5(1)]. Equivalently,

$$\operatorname{fib}(A' \longrightarrow T'_n) \simeq (J')^{\otimes_{A'}(n+1)}.$$

Compatibility of the standard t-structure with tensor products places the right-hand side in $(\mathrm{Mod}_{A'})_{\geq n+1}$. Its inverse limit over n therefore vanishes, and left completeness gives

$$A' \simeq \operatorname*{Tot}_{\Delta}(B')^{\otimes_{A'}(\bullet+1)}.$$

Since A' was arbitrary, the Amitsur equivalence is universal. This is also the base-change assertion of [10, Remark 2.3.7(2)]. $\square$

Lemma 2.5 (The constant Postnikov hypercover)**.** *Let A be a connective E_∞-ring. The constant simplicial affine* $\operatorname{Spec} H\pi_0 A$*, augmented to* $\operatorname{Spec} A$*, is a hypercover for the canonical topology. In particular, the constant simplicial affine* $\operatorname{Spec} H\mathbb{Z}$ *is a canonical hypercover of* $\operatorname{Spec} \mathbb{S}$*.*

Proof. Let M_n be the nth matching affine of the augmented constant simplicial object and write C_n for its coordinate ring. For $n = 0$ one has $C_0 = A$; for $n > 0$ the matching category is finite, so C_n is a finite colimit of copies of $H\pi_0 A$ over A. After applying the colimit-preserving functor $\pi_0 : \mathrm{CAlg}^{\mathrm{cn}} \to \mathrm{CRing}$, the augmented diagram becomes the constant identity diagram on $\pi_0 A$. Thus the matching map $C_n \to H\pi_0 A$ is an isomorphism on π_0. Since its target is 0-truncated and its source is connective, its fiber is 1-connective. Hence Lemma 2.4 says that the corresponding map $\operatorname{Spec} H\pi_0 A \to M_n$ is a canonical cover. This holds for every n, including $n = 0$, and proves the hypercover assertion. Apply it to $\mathbb{S} \to H\mathbb{Z}$ for the final statement. $\square$

Lemma 2.6 (Coconnective sheaves and Postnikov hypercovers)**.** *Let*

$$F : \mathrm{CAlg}^{\mathrm{cn}} \longrightarrow \mathrm{Sp}$$

be a spectrum-valued presheaf on connective spectral affines, written in ring coordinates.

1. *If F is a canonical hypersheaf, then for every connective E_∞-ring A the map*

$$F(A) \longrightarrow F(H\pi_0 A)$$

is an equivalence.

2. *Suppose that F is a sheaf for the canonical topology and that there is an integer d such that $F(A) \in \mathrm{Sp}_{\leq d}$ for every connective A. Then F is a canonical hypersheaf, and the conclusion of (1) holds.*

Proof. For (1), apply hyperdescent to the constant hypercover of Lemma 2.5. Its value under F is the constant cosimplicial spectrum on $F(H\pi_0 A)$, whose totalization is $F(H\pi_0 A)$.

For (2), regard F as an object of the stable category of spectrum-valued canonical sheaves. For every integer m, the space-valued sheaf

$$\Omega^\infty \Sigma^m F$$

is pointwise $(d+m)$-truncated when $d+m \geq 0$ and pointwise 0-truncated when $d+m < 0$. Truncatedness in the canonical ∞-topos can be tested on the representable generators, so these are truncated objects of that ∞-topos. Every truncated object of an ∞-topos is hypercomplete [16, Lemma 6.5.2.9]. Hence, for every canonical hypercover $U_\bullet \to X$, applying $\Omega^\infty\Sigma^m$ to

$$F(X) \longrightarrow \operatorname*{Tot}_{[q]\in\Delta} F(U_q)$$

gives an equivalence for every m. The functors $\{\Omega^\infty\Sigma^m\}_{m\in\mathbb{Z}}$ preserve limits and are jointly conservative on spectra. The displayed map is therefore an equivalence. This is also the spectrum-sheaf hypercompleteness criterion of [17, Proposition 1.3.3.3]. Thus F is a hypersheaf, and (1) applies. □

2.2 Module and finite-group descent

Lemma 2.7 (Bounded-below module descent)**.** *Let $\{A \to B_i\}_{i\in I}$ be canonical. If M is a bounded-below A-module, then*

$$M \simeq \operatorname*{Tot}_q \prod_{\alpha\in I^{q+1}} (M \otimes_A B_\alpha). \tag{2.3}$$

The equivalence is natural in M. It also holds for a bounded-below equivariant module over a canonical family in $\mathrm{CAlg}^{\mathrm{cn}}(\mathrm{Fun}(BG, \mathrm{Sp}))$*.*

Proof. Let T denote the right side of (2.3), formed in the unbounded stable module category. For every sufficiently large r, apply (2.1) to the connective square-zero algebra $A \oplus \Sigma^r M$. Taking the square-zero ideal of the resulting equivalence gives

$$\Sigma^r M \simeq \tau_{\geq 0}(\Sigma^r T), \qquad \text{hence} \qquad \tau_{\geq -r} M \simeq \tau_{\geq -r} T.$$

This holds for arbitrarily large r, so Postnikov separatedness gives $M \simeq T$. In particular, the proof never moves a tensor product through an infinite product. The same square-zero argument works in equivariant commutative algebras, where limits and equivalences are detected on underlying modules. □

Lemma 2.8 (Finite-group Tate descent)**.** *Let G be finite and let $M \in \mathrm{Fun}(BG, \mathrm{Mod}_A)$ be bounded below. For a canonical family, the natural maps*

$$M^{hG} \longrightarrow \operatorname*{Tot}_q \prod_\alpha (M \otimes_A B_\alpha)^{hG}, \qquad M_{hG} \longrightarrow \operatorname*{Tot}_q \prod_\alpha (M \otimes_A B_\alpha)_{hG}$$

are equivalences. Consequently the analogous Tate comparison is an equivalence. If $G \lhd H$, all three comparisons retain their residual H/G-actions.

Proof. Homotopy fixed points and totalization are limits, so the fixed-point claim follows from Lemma 2.7 in G-objects. On the orbit side, M_{hG} is bounded below and

$$(M \otimes_A B_\alpha)_{hG} \simeq M_{hG} \otimes_A B_\alpha,$$

because B_α has trivial action. Apply Lemma 2.7 to M_{hG}. Exactness of totalization and the norm cofiber sequence give the Tate assertion. These constructions take place in residual H/G-objects when an H-action is present. □

2.3 Split contractions

Lemma 2.9 (Split-member contraction)**.** *Let $\mathcal{C}$ be an infinity-category with pushouts, let $\mathcal{D}$ admit small products and totalizations, and let $\Phi : \mathcal{C} \to \mathcal{D}$ be a functor. Given a small family $\{u_i : X \to X_i\}_{i \in I}$, put*

$$X_{i_0,\dots,i_q} = X_{i_0} \amalg_X \cdots \amalg_X X_{i_q}, \qquad C^q_\Phi = \prod_{(i_0,\dots,i_q) \in I^{q+1}} \Phi(X_{i_0,\dots,i_q}).$$

The canonical maps between iterated pushouts make $\Phi(X) \to C^\bullet_\Phi$ an augmented cosimplicial object. If one u_{i_*} admits a retraction $r : X_{i_*} \to X$ in $\mathcal{C}$, then this augmentation is split. In particular, the augmentation induces an equivalence

$$\Phi(X) \xrightarrow{\sim} \mathrm{Tot}(C^\bullet_\Phi).$$

Proof. For a factor of C^{q-1}_Φ indexed by $\mathbf{i} = (i_0, \dots, i_{q-1})$, project C^q_Φ to the factor indexed by $(i_*, \mathbf{i})$ and apply Φ to the map

$$X_{i_*,i_0,\dots,i_{q-1}} \longrightarrow X_{i_0,\dots,i_{q-1}}$$

induced by r on the first factor and the identity on the remaining factors. These maps, together with the analogous map in cosimplicial degree zero, define an extra codegeneracy. The extra-codegeneracy identities reduce to $r \circ u_{i_*} = \mathrm{id}_X$ and the associativity of pushouts. Hence the augmentation is split. A split augmented cosimplicial object is a limit diagram, which gives the displayed equivalence. The construction uses only projections from the indicated products, so no finiteness assumption on I is needed. □

3 Cotangent-polynomial descent

Cotangent descent is the main nonlinear input of the paper. For a map of pairs, transitivity separates the desired base-change term from a relative error. The binomial filtration reduces descent to these pieces: module descent treats the base-change term, while a second Čech direction makes every positive error term split. A pair (R/A) means a map $A \to R$.

Antieau records, with attribution to joint work with Iwasa and Krause, the singleton canonical-cover form of Hodge-completed infinitesimal descent and its bounded-below cotangent symmetric-power consequence in [1, Theorem 2.11 and Corollary 2.12]. We prove the small-family version with arbitrary bounded-below coefficients and record the finite-group Tate refinement needed below.

3.1 Complete filtrations and arbitrary Čech limits

For a stable infinity-category $\mathcal{C}$ with small limits, write

$$\mathrm{Fil}(\mathcal{C}) = \mathrm{Fun}(\mathbb{N}^{\mathrm{op}}, \mathcal{C})$$

for decreasing nonnegative filtrations, and write

$$\mathrm{Fil}_{\mathbb{Z}}(\mathcal{C}) = \mathrm{Fun}(\mathbb{Z}^{\mathrm{op}}, \mathcal{C})$$

for decreasing $\mathbb{Z}$-indexed filtrations. Thus an object has maps $F^{n+1}X \to F^n X$. In either category put

$$\mathrm{gr}^n X = \mathrm{cofib}(F^{n+1}X \to F^n X), \qquad F^\infty X = \lim_{n\to+\infty} F^n X,$$

and let $\mathrm{Fil}^\wedge(\mathcal{C})$ and $\mathrm{Fil}^\wedge_{\mathbb{Z}}(\mathcal{C})$ denote the full subcategories of derived complete filtrations, defined by $F^\infty X \simeq 0$.

Lemma 3.1 (Graded detection of complete filtered limits)**.** *The subcategories* $\mathrm{Fil}^\wedge(\mathcal{C})$ *and* $\mathrm{Fil}^\wedge_{\mathbb{Z}}(\mathcal{C})$ *are closed under all small limits. For either indexing category, every functor*

$$\mathrm{gr}^n : \mathrm{Fil}^\wedge(\mathcal{C}) \longrightarrow \mathcal{C} \quad (n \geq 0), \qquad \mathrm{gr}^n : \mathrm{Fil}^\wedge_{\mathbb{Z}}(\mathcal{C}) \longrightarrow \mathcal{C} \quad (n \in \mathbb{Z})$$

preserves all small limits, and the family of all available graded-piece functors is jointly conservative. Consequently, a cone of complete filtered objects is a limit cone as soon as it is a limit cone on every associated graded piece.

Proof. Limits of filtrations are computed weightwise. If $D : K \to \mathrm{Fil}(\mathcal{C})$ or $D : K \to \mathrm{Fil}_{\mathbb{Z}}(\mathcal{C})$ is a diagram of complete filtrations, Fubini for limits gives

$$\lim_{n\to+\infty} \lim_{k\in K} F^n D_k \simeq \lim_{k\in K} \lim_{n\to+\infty} F^n D_k \simeq 0.$$

Thus the limit is complete. In a stable category a cofiber is a suspended fiber, so the cofiber functor on arrows preserves limits. Hence

$$\mathrm{gr}^n\left(\lim_{k\in K} D_k\right) \simeq \lim_{k\in K} \mathrm{gr}^n(D_k).$$

For conservativity, let $f : X \to Y$ be a graded equivalence between complete filtrations and let $H = \mathrm{fib}(f)$. Then H is complete, every $\mathrm{gr}^n H$ is zero, and therefore every transition $F^{n+1}H \to F^n H$ is an equivalence. For fixed n the tail is constant up to equivalence, whence

$$F^n H \simeq \lim_{r\geq n} F^r H \simeq F^\infty H \simeq 0.$$

Thus f is an equivalence. Applying this observation to the comparison from the cone point to the limit proves the final assertion. $\square$

The arbitrary-limit statement applies in particular to products, totalizations, and filtration limits. No Mittag–Leffler or uniform-connectivity hypothesis is needed, since all three constructions are limits. Rapid connectivity enters later only because finite-group Tate does not preserve arbitrary inverse limits. Completeness itself is essential; a nonzero constant filtration has zero associated graded in every weight.

Proposition 3.2 (Cotangent polynomials and Tate descent)**.** *For every $i \in I$, let*

$$\begin{array}{ccc} A & \longrightarrow & B_i \\ \downarrow & & \downarrow \\ R & \longrightarrow & S_i \end{array}$$

be a commutative square of connective E_∞-rings, and suppose that the small family $\{R \to S_i\}_{i\in I}$ is canonical. For $\alpha = (i_0, \dots, i_q)$ put

$$B_\alpha = B_{i_0} \otimes_A \cdots \otimes_A B_{i_q}, \qquad S_\alpha = S_{i_0} \otimes_R \cdots \otimes_R S_{i_q}.$$

Let M be a bounded-below A-module and let V be a bounded-below spectrum, possibly with an action of a group G, while all rings have trivial action. For every $d \geq 0$, the natural map

$$\begin{aligned} &(M \otimes_A R) \otimes_R \mathrm{Sym}^d_R(\mathbb{L}^{E_\infty}_{R/A} \otimes V) \\ &\qquad \longrightarrow \operatorname*{Tot}_q \prod_{\alpha \in I^{q+1}} (M \otimes_A S_\alpha) \otimes_{S_\alpha} \mathrm{Sym}^d_{S_\alpha}(\mathbb{L}^{E_\infty}_{S_\alpha/B_\alpha} \otimes V) \end{aligned} \tag{3.1}$$

is a G-equivariant equivalence. If G is finite, the map from the Tate construction of the source to the totalization of the termwise Tate constructions on the right is also an equivalence. If the data are H-equivariant and $G \lhd H$, the Tate comparison retains the residual H/G-action.

Proof. The transitivity filtration. Set

$$C_\alpha = R \otimes_A B_\alpha, \qquad Q_\alpha = \mathbb{L}^{E_\infty}_{S_\alpha/C_\alpha} \otimes V.$$

Base change and transitivity for the E_∞ cotangent complex [15, Corollary 7.3.3.6 and Proposition 7.3.3.7] give a natural cofiber sequence

$$S_\alpha \otimes_R (\mathbb{L}^{E_\infty}_{R/A} \otimes V) \longrightarrow \mathbb{L}^{E_\infty}_{S_\alpha/B_\alpha} \otimes V \longrightarrow Q_\alpha. \tag{3.2}$$

The derived binomial filtration of the dth symmetric power in (3.2) is finite. Its graded pieces are indexed by the number b of factors contributed by Q_α.

The base-change piece. The piece with $b = 0$ is the base change of the source of (3.1), so Lemma 2.7 proves descent for this piece.

The positive error pieces. Fix $b > 0$. After collecting the factors independent of α, the corresponding graded term has the form

$$E_{\alpha,b} = (N_b \otimes_R S_\alpha) \otimes_{S_\alpha} \mathrm{Sym}^b_{S_\alpha}(Q_\alpha),$$

where N_b is a bounded-below G-R-module. Here we use that cotangent complexes of connective E_∞-rings are connective [15, Proposition 7.4.3.9]. Apply module descent in a second Čech direction, indexed by $\beta \in I^{p+1}$. Fubini for small limits gives

$$\operatorname*{Tot}_q \prod_\alpha E_{\alpha,b} \simeq \operatorname*{Tot}_p \operatorname*{Tot}_q \prod_{\beta,\alpha} (E_{\alpha,b} \otimes_R S_\beta). \tag{3.3}$$

For fixed $\beta = (j_0, \dots, j_p)$, put

$$T_{\beta,\alpha} = S_\beta \otimes_R S_\alpha, \qquad D_{\beta,\alpha} = S_\beta \otimes_A B_\alpha.$$

Base change identifies the relative factor in $E_{\alpha,b} \otimes_R S_\beta$ with

$$\mathbb{L}^{E_\infty}_{T_{\beta,\alpha}/D_{\beta,\alpha}} \otimes V.$$

As α varies, these are the family Amitsur terms for the map of pairs

$$(S_\beta/S_\beta) \longrightarrow \{(S_\beta \otimes_R S_i)/(S_\beta \otimes_A B_i)\}_{i\in I}.$$

Colimits in $\mathrm{Fun}(\Delta^1, \mathrm{CAlg}^{\mathrm{cn}})$ are computed objectwise, so its iterated pushouts are precisely the pairs $(T_{\beta,\alpha}/D_{\beta,\alpha})$ above. The member $i = j_0$ retracts by multiplying the repeated S_{j_0} factor; on the denominator one first uses $B_{j_0} \to S_{j_0}$. The augmented object is therefore split in the infinity-category $\mathrm{Fun}(\Delta^1, \mathrm{CAlg}^{\mathrm{cn}})$ of pairs. On the undercategory of (S_β/S_β), apply Lemma 2.9 to the functor

$$\Phi_{\beta,b} : \mathrm{Fun}(\Delta^1, \mathrm{CAlg}^{\mathrm{cn}})_{(S_\beta/S_\beta)/} \longrightarrow \mathrm{Fun}(BG, \mathrm{Sp}),$$
$$\Phi_{\beta,b}(T/D) = (N_b \otimes_R T) \otimes_T \mathrm{Sym}^b_T(\mathbb{L}^{E_\infty}_{T/D} \otimes V).$$

Its value at the augmentation is zero, because $\mathbb{L}^{E_\infty}_{S_\beta/S_\beta} = 0$ and $b > 0$. The lemma therefore identifies the α-totalization in (3.3) with zero. Thus every positive error piece vanishes, and induction through the finite binomial filtration proves (3.1).

The Tate comparison. For the Tate statement, use Lemma 2.8 on the piece with $b = 0$. Fix $b > 0$. Since every $E_{\alpha,b}$ is a bounded-below G-R-module, a second application of Lemma 2.8, now in the β-direction, and Fubini for limits give

$$\operatorname*{Tot}_q \prod_\alpha E^{tG}_{\alpha,b} \simeq \operatorname*{Tot}_q \operatorname*{Tot}_p \prod_{\alpha,\beta} (E_{\alpha,b} \otimes_R S_\beta)^{tG} \simeq \operatorname*{Tot}_p \operatorname*{Tot}_q \prod_{\beta,\alpha} (E_{\alpha,b} \otimes_R S_\beta)^{tG}.$$

For fixed β, the augmented α-object constructed above is split with zero value at its augmentation. It remains split after applying $(-)^{tG}$, so the inner α-totalization in the last expression is zero. Exactness of Tate and totalization, followed by the same finite filtration argument, proves the claim. All constructions preserve the stated residual action. □

The case $d = 1$ gives descent for the relative cotangent complex itself. The full family of symmetric powers also reconstructs a naturally occurring complete filtered theory.

Corollary 3.3 (Hodge-completed descent)**.** *Under the hypotheses of Proposition 3.2, Hodge-completed E_∞-infinitesimal cohomology satisfies filtered ordinary Čech descent. More precisely, there is an equivalence*

$$F^\bullet_{\mathrm{H}} \widehat{\mathrm{Inf}}^{E_\infty}_{R/A} \simeq \operatorname*{Tot}_q \prod_{\alpha\in I^{q+1}} F^\bullet_{\mathrm{H}} \widehat{\mathrm{Inf}}^{E_\infty}_{S_\alpha/B_\alpha} \tag{3.4}$$

of complete filtered E_∞-algebras, and hence of their underlying complete filtered spectra.

Proof. Antieau's construction has a complete Hodge filtration, and completion is taken along this same filtration. Its associated graded is

$$\mathrm{gr}^d_{\mathrm{H}} \widehat{\mathrm{Inf}}^{E_\infty}_{R/A} \simeq \mathrm{Sym}^d_R(\mathbb{L}^{E_\infty}_{R/A}[-1]); \tag{3.5}$$

see [1, Lemma 2.2]. Apply Proposition 3.2 with $M = A$ and $V = \Sigma^{-1}\mathbb{S}$. It identifies every associated graded piece of the two sides of (3.4). Lemma 3.1 proves the filtered equivalence. Its multiplicative structure is natural throughout the construction. □

4 From cotangent descent to cyclotomic descent

Cotangent-polynomial descent reaches THH through the circle-equivariant HKR filtration. This is the route used for cyclotomic descent, because it survives finite-group Tate constructions. The underlying circle-equivariant theorem also has a shorter proof from the circle tensor; we include it as an alternative.

All circle and finite-group actions in this paper are ordinary (naive) actions: an action of G means an object of $\mathrm{Fun}(BG, -)$.

4.1 Equivariant THH descent

Let

$$V_{S^1} = \mathrm{fib}(\Sigma^\infty_+ S^1 \longrightarrow \mathbb{S}),$$

where the source has its translation action and the target has the trivial circle action. Its underlying spectrum is $\Sigma\mathbb{S}$, but its circle action is not trivial. We write $\mathbb{L}_A = \mathbb{L}^{E_\infty}_{A/\mathbb{S}}$.

Lemma 4.1 (The equivariant copower cotangent complex)**.** *Put $C_A = S^1 \otimes A = \mathrm{THH}(A)$. Give C_A the rotation action and A the trivial circle action, and let $\varepsilon : C_A \to A$ be the collapse augmentation. There is a natural equivalence*

$$A \otimes_{C_A} \mathbb{L}^{E_\infty}_{C_A} \simeq \mathbb{L}^{E_\infty}_A \otimes \Sigma^\infty_+ S^1 \quad \textit{in } \mathrm{Fun}(B\mathbb{T}, \mathrm{Mod}_A),$$

under which the first map in the transitivity triangle for $C_A \to A$ is

$$\mathbb{L}^{E_\infty}_A \otimes \Sigma^\infty_+ S^1 \xrightarrow{1\otimes \mathrm{coll}} \mathbb{L}^{E_\infty}_A.$$

Consequently,

$$\mathbb{L}^{E_\infty}_{A/C_A}[-1] \simeq \mathbb{L}^{E_\infty}_A \otimes V_{S^1} \quad \textit{in } \mathrm{Fun}(B\mathbb{T}, \mathrm{Mod}_A).$$

Proof. For an A-module M, viewed with trivial circle action, the universal property of the cotangent complex and the copower–mapping adjunction give natural equivalences

$$\begin{aligned}\mathrm{map}_A(A \otimes_{C_A} \mathbb{L}^{E_\infty}_{C_A}, M) &\simeq \mathrm{Der}_\varepsilon(C_A, M)\\ &\simeq \mathrm{map}_{\mathrm{Sp}}(\Sigma^\infty_+ S^1, \mathrm{Der}_{\mathrm{id}_A}(A, M))\\ &\simeq \mathrm{map}_A(\mathbb{L}^{E_\infty}_A \otimes \Sigma^\infty_+ S^1, M).\end{aligned}$$

The same calculation with an arbitrary equivariant A-module, using equivariant mapping spectra, identifies the representing map in $\mathrm{Fun}(B\mathbb{T}, \mathrm{Mod}_A)$. Equivalently, the underlying equivalence constructed above is natural for rotation and is therefore an equivariant equivalence. The middle equivalence is the fiber over the constant map at id_A in

$$\mathrm{Map}_{\mathrm{CAlg}}(S^1 \otimes A, A \oplus M) \simeq \mathrm{Map}(S^1, \mathrm{Map}_{\mathrm{CAlg}}(A, A \oplus M)).$$

This construction is natural for rotations of S^1, so the representing equivalence is circle-equivariant. Naturality for the collapse $S^1 \to *$ identifies the induced map to $\mathbb{L}^{E_\infty}_A$ with $1 \otimes \mathrm{coll}$. The transitivity triangle

$$A \otimes_{C_A} \mathbb{L}^{E_\infty}_{C_A} \longrightarrow \mathbb{L}^{E_\infty}_A \longrightarrow \mathbb{L}^{E_\infty}_{A/C_A}$$

is circle-equivariant. Taking the fiber of its first map proves the final claim. $\square$

Proposition 4.2 (Circle-equivariant HKR filtration)**.** *The complete circle-equivariant HKR filtration on* $\mathrm{THH}(A)$ *has graded pieces*

$$\mathrm{gr}^d_{\mathrm{HKR}}\,\mathrm{THH}(A) \simeq \mathrm{Sym}^d_A(\mathbb{L}^{E_\infty}_A \otimes V_{S^1}) \quad \textit{in } \mathrm{Fun}(B\mathbb{T}, \mathrm{Mod}_A). \tag{4.1}$$

If $F^\bullet_{\mathrm{HKR}}$ *denotes this filtration, its tails* $F^n_{\mathrm{HKR}}\mathrm{THH}(A)$ *become increasingly connective, uniformly after connective base change.*

Proof. Antieau's HKR filtration is the Hodge filtration

$$F^\bullet_{\mathrm{HKR}}\mathrm{THH}(A) = F^\bullet_{\mathrm{H}}\,\mathrm{Inf}^{E_\infty}_{A/C_A}.$$

It is functorial in the pair $C_A \to A$ by [1, Construction 2.1 and Proposition 2.6]. The rotation action on C_A, together with the trivial action on A, therefore equips the filtered object with an ordinary circle action; see also [1, Remark 4.3]. The natural associated-graded equivalence of [1, Lemma 2.2], applied in circle objects, and Lemma 4.1 give

$$\begin{aligned}\mathrm{gr}^d_{\mathrm{HKR}}\,\mathrm{THH}(A) &\simeq \mathrm{Sym}^d_A(\mathbb{L}^{E_\infty}_{A/C_A}[-1]) \\ &\simeq \mathrm{Sym}^d_A(\mathbb{L}^{E_\infty}_A \otimes V_{S^1})\end{aligned}$$

in $\mathrm{Fun}(B\mathbb{T}, \mathrm{Mod}_A)$. After forgetting the action, $V_{S^1} \simeq \Sigma\mathbb{S}$.

Completeness is [1, Proposition 4.1]. Since $\mathbb{L}^{E_\infty}_A$ is connective, the dth graded piece is d-connective. For $N > n$, put

$$Q_{n,N} = \mathrm{cofib}\big(F^N_{\mathrm{HKR}}\mathrm{THH}(A) \longrightarrow F^n_{\mathrm{HKR}}\mathrm{THH}(A)\big).$$

This is a finite extension of the graded pieces with $n \le d < N$ and is therefore n-connective. In each fixed homotopy degree the inverse system $\{Q_{n,N}\}_N$ is eventually constant, so its Milnor $\lim^1$ vanishes. Completeness gives

$$F^n_{\mathrm{HKR}}\mathrm{THH}(A) \simeq \lim_{N>n} Q_{n,N} \in \mathrm{Sp}_{\ge n}.$$

The estimate depends only on n, and hence holds simultaneously for all connective rings in a canonical-family Čech diagram. ☐

Theorem 4.3 (Complete filtered equivariant THH descent)**.** *For every small canonical family* $\{A \to B_i\}_{i\in I}$*, the natural map*

$$F^\bullet_{\mathrm{HKR}}\mathrm{THH}(A) \longrightarrow \operatorname*{Tot}_q \prod_{\alpha\in I^{q+1}} F^\bullet_{\mathrm{HKR}}\mathrm{THH}(B_\alpha) \tag{4.2}$$

is an equivalence in $\mathrm{Fil}^\wedge(\mathrm{Fun}(B\mathbb{T}, \mathrm{Sp}))$*. It respects the functorial filtered* E_∞*-multiplication and ordinary circle action. In particular, evaluation at filtration degree zero gives an equivalence*

$$\mathrm{THH}(A) \longrightarrow \operatorname*{Tot}_q \prod_{\alpha\in I^{q+1}} \mathrm{THH}(B_\alpha) \tag{4.3}$$

in $\mathrm{Fun}(B\mathbb{T}, \mathrm{Sp})$*.*

Proof from cotangent descent. Apply Proposition 3.2 to the squares whose upper row is $\mathbb{S} \to \mathbb{S}$ and whose lower row is $A \to B_i$, with $M = \mathbb{S}$ and $V = V_{S^1}$. By (4.1), every HKR graded piece satisfies equivariant family descent. Proposition 4.2 gives completeness at every vertex, while products and totalization preserve complete filtrations. Lemma 3.1 therefore proves (4.2). Functoriality of the HKR construction preserves its multiplication and ordinary circle action. Since $F^0_{\mathrm{HKR}}\mathrm{THH}(T) = \mathrm{THH}(T)$ and evaluation at degree zero preserves limits, the unfiltered equivalence (4.3) follows. ☐

Alternative proof of the unfiltered equivalence: the circle tensor. The augmentation

$$\varepsilon_A : \mathrm{THH}(A) = S^1 \otimes A \longrightarrow A$$

has a nonequivariant section induced by the basepoint of S^1. It is a π_0-isomorphism, and the section makes its fiber connective; hence that fiber is 1-connective. The proof of Lemma 2.4, carried out in circle objects, shows that ε_A is canonical equivariantly.

Compose it with the canonical family $\{A \to B_i\}$. The resulting family $\{\mathrm{THH}(A) \to B_i\}$ factors through $\{\mathrm{THH}(A) \to \mathrm{THH}(B_i)\}$. The refinement clause of Lemma 2.3 makes the latter family canonical in circle-equivariant commutative algebras. Since $S^1 \otimes -$ preserves pushouts,

$$\mathrm{THH}(B_{i_0}) \otimes_{\mathrm{THH}(A)} \cdots \otimes_{\mathrm{THH}(A)} \mathrm{THH}(B_{i_q}) \simeq \mathrm{THH}(B_\alpha).$$

The Amitsur equivalence is initially formed in connective commutative algebras. Applying Lemma 2.7 to the unit module over $\mathrm{THH}(A)$ identifies the same infinite totalization in the ambient stable category of circle spectra, giving (4.3). □

4.2 Tate continuity and cyclotomic descent

Tate constructions do not commute with arbitrary inverse limits. The HKR tower is usable because its tails become uniformly connective. The next two lemmas isolate this continuity argument.

Lemma 4.4 (Rapid complete filtrations)**.** *Let $F^\bullet V$ be a complete decreasing filtration. If $\mathrm{gr}^i V$ is $(i+c)$-connective, then $F^n V$ is $(n+c)$-connective. The assertion holds uniformly in a diagram whenever the bound on the graded pieces is uniform.*

Proof. For $N > n$, let $Q_{n,N} = \mathrm{cofib}(F^N V \to F^n V)$. It is a finite extension of the graded pieces with indices $n \leq i < N$, and is therefore $(n+c)$-connective. For each homotopy group the tower stabilizes as $N \to \infty$, so its $\lim^1$ vanishes. Completeness identifies $F^n V$ with $\lim_N Q_{n,N}$ and gives the stated bound. □

Lemma 4.5 (Tate continuity for the HKR tower)**.** *Let $V \simeq \lim_n V/F^n V$ be a complete tower of bounded-below C_m-spectra. Suppose the connectivity of $F^n V$ tends uniformly to $+\infty$. Then*

$$V^{tC_m} \simeq \lim_n (V/F^n V)^{tC_m}.$$

Proof. Homotopy orbits are exact and preserve lower connectivity for a finite group, so Postnikov completeness gives $V_{hC_m} \simeq \lim_n (V/F^n V)_{hC_m}$. Homotopy fixed points commute with all limits. Taking inverse limits in the norm cofiber sequences proves the claim. □

Corollary 4.6 (Complete filtered finite-group Tate descent)**.** *For every finite subgroup $C_m \subset \mathbb{T}$, put*

$$F^n_{\mathrm{HKR},tC_m} \mathrm{THH}(T) = \left(F^n_{\mathrm{HKR}} \mathrm{THH}(T)\right)^{tC_m}.$$

This is a complete filtration on $\mathrm{THH}(T)^{tC_m}$, with its residual $\mathbb{T}/C_m$-action, and

$$\mathrm{gr}^n_{\mathrm{HKR},tC_m} \mathrm{THH}(T) \simeq \left(\mathrm{gr}^n_{\mathrm{HKR}} \mathrm{THH}(T)\right)^{tC_m}.$$

For every small canonical family the natural map

$$F^\bullet_{\mathrm{HKR},tC_m} \mathrm{THH}(A) \longrightarrow \operatorname*{Tot}_q \prod_{\alpha \in I^{q+1}} F^\bullet_{\mathrm{HKR},tC_m} \mathrm{THH}(B_\alpha)$$

is an equivalence in $\mathrm{Fil}^\wedge(\mathrm{Fun}(B(\mathbb{T}/C_m), \mathrm{Sp}))$.

Proof. Exactness of finite-group Tate gives the associated-graded formula. Proposition 4.2 and Lemma 4.5 identify

$$\mathrm{THH}(T)^{tC_m} \simeq \lim_n \big(\mathrm{THH}(T)/F^n_{\mathrm{HKR}}\mathrm{THH}(T)\big)^{tC_m}.$$

Taking the fiber of the comparison with the nth quotient and then the limit over n shows $\lim_n (F^n_{\mathrm{HKR}}\mathrm{THH}(T))^{tC_m} \simeq 0$; hence the displayed Tate filtration is complete. The Tate clause of Proposition 3.2 gives canonical-family descent for every graded piece. Apply Lemma 3.1. ☐

Remark 4.7 (Filtered cyclotomic compatibility). Corollary 4.6 concerns the termwise Tate filtration of an ordinary circle-equivariant filtered object. It does not place the HKR filtration in CycSp. In particular, naturality of the unfiltered cyclotomic Frobenius does not produce maps

$$F^n_{\mathrm{HKR}}\mathrm{THH}(T) \longrightarrow \big(F^{pn}_{\mathrm{HKR}}\mathrm{THH}(T)\big)^{tC_p}$$

or a weight-preserving variant. The proof of cyclotomic descent below uses the filtered Tate comparison only after evaluating the complete filtration at degree zero.

Theorem 4.8 (Tate and cyclotomic descent)**.** *For every small canonical family and every prime p, there is a $\mathbb{T}/C_p$-equivariant equivalence*

$$\mathrm{THH}(A)^{tC_p} \simeq \operatorname*{Tot}_q \prod_{\alpha\in I^{q+1}} \mathrm{THH}(B_\alpha)^{tC_p}. \tag{4.4}$$

Together with Theorem 4.3, these comparisons identify the Čech limit in Nikolaus–Scholze cyclotomic spectra. Consequently

$$\mathrm{THH}(A) \simeq \operatorname*{Tot}_q \prod_\alpha \mathrm{THH}(B_\alpha) \quad \textit{in } \mathrm{CycSp}, \qquad \mathrm{TC}(A) \simeq \operatorname*{Tot}_q \prod_\alpha \mathrm{TC}(B_\alpha).$$

Proof. For $n \geq 0$ set

$$Q_n(T) = \mathrm{cofib}\big(F^n_{\mathrm{HKR}}\mathrm{THH}(T) \longrightarrow \mathrm{THH}(T)\big).$$

This quotient has a finite filtration with graded pieces $\mathrm{gr}^d_{\mathrm{HKR}}\,\mathrm{THH}(T)$ for $0 \leq d < n$. For a finite subgroup $C_m \subset \mathbb{T}$, the Tate clause of Proposition 3.2 gives canonical-family descent after C_m-Tate for each of these pieces. Exactness of $(-)^{tC_m}$ and induction through the finite filtration therefore give descent for $Q_n(-)^{tC_m}$. Proposition 4.2 and Lemma 4.4 give the uniform connectivity required by Lemma 4.5. Hence, for every prime p,

$$\begin{aligned}
\mathrm{THH}(A)^{tC_p} &\simeq \lim_n Q_n(A)^{tC_p} \\
&\simeq \lim_n \operatorname*{Tot}_q \prod_\alpha Q_n(B_\alpha)^{tC_p} \\
&\simeq \operatorname*{Tot}_q \prod_\alpha \lim_n Q_n(B_\alpha)^{tC_p} \simeq \operatorname*{Tot}_q \prod_\alpha \mathrm{THH}(B_\alpha)^{tC_p}.
\end{aligned}$$

These equivalences are natural in the ring and retain the residual $\mathbb{T}/C_p$-action. No cyclotomic structure on the finite quotients Q_n is used.

It remains to promote the underlying and Tate comparisons to a limit in cyclotomic spectra. Let J be the small indexing category for the whole family Čech limit, and let $D : J \to \mathrm{CycSp}$ have entries $D_{q,\alpha} = \mathrm{THH}(B_\alpha)$, with the cyclotomic structure of [20, Example II.1.2(i)]. Thus

$$\lim_{j\in J} UD_j = \operatorname*{Tot}_q \prod_{\alpha\in I^{q+1}} U\mathrm{THH}(B_\alpha),$$

where $U : \mathrm{CycSp} \to \mathrm{Fun}(B\mathbb{T}, \mathrm{Sp})$ is forgetful. Put

$$\mathcal{C} = \mathrm{Fun}(B\mathbb{T}, \mathrm{Sp}), \qquad \mathcal{D} = \prod_{p \in \mathcal{P}} \mathcal{C},$$

and define $F(X) = (X)_p$ and $G(X) = (X^{tC_p})_p$, using the pth-power identification $\mathbb{T}/C_p \simeq \mathbb{T}$. By [20, Definition II.1.6],

$$\mathrm{CycSp} \simeq \mathrm{LEq}(F, G).$$

Theorem 4.3 identifies the underlying cone

$$U\mathrm{THH}(A) \xrightarrow{\sim} \lim_{j \in J} UD_j.$$

For every prime p, equation (4.4) identifies the canonical Tate comparison

$$(U\mathrm{THH}(A))^{tC_p} \xrightarrow{\sim} \lim_{j \in J} (UD_j)^{tC_p}.$$

Taking the product over all primes says exactly that G preserves this particular J-limit; no preservation of arbitrary limits by Tate is being asserted.

The diagonal functor $F : \mathcal{C} \to \prod_p \mathcal{C}$ preserves all limits, while the preceding Tate comparison proves preservation of this particular J-limit by G.

The comparisons respect the cyclotomic structure maps. For every prime p, naturality in the ring of the underlying and Tate comparisons, together with functoriality of the cyclotomic Frobenius on THH, gives the commutative square

$$\begin{tikzcd} U\mathrm{THH}(A) \arrow[r] \arrow[d, "\varphi_p"'] & \lim_{j \in J} UD_j \arrow[d, "\lim_{j \in J} \varphi_{p,j}"] \\ (U\mathrm{THH}(A))^{tC_p} \arrow[r] & \lim_{j \in J} (UD_j)^{tC_p}. \end{tikzcd}$$

All finite-quotient and inverse-limit comparisons used for the bottom arrow are natural in the ring. The cone is therefore a cone in $\mathrm{LEq}(F, G)$. The limit criterion [20, Proposition II.1.5(v)] now shows that D has the displayed limit in CycSp and that U preserves it. Equivalences in the lax equalizer are detected on underlying objects by [20, Proposition II.1.5(ii)], which proves the asserted equivalence in CycSp.

Finally, [20, Definition II.1.8] identifies

$$\mathrm{TC}(V) = \mathrm{map}_{\mathrm{CycSp}}(\mathbb{S}_{\mathrm{cyc}}, V).$$

This representable mapping-spectrum functor preserves limits, giving the stated descent equivalence for TC. ☐

5 Filtered descent and the HRW obstruction

The preceding sections provide the Hodge, HKR, finite-group Tate, and cyclotomic descent statements. We first establish the remaining p-adic filtered statement and then collect the results before studying the Hahn–Raksit–Wilson even filtration.

5.1 A universal filtered TC theorem

The motivic filtration is not the only natural complete filtration on TC. The arithmetic p-adic filtration gives a filtered theorem for all connective spectral rings and all canonical families.

Theorem 5.1 (p-adic filtered TC descent)**.** *Fix a prime p and write*

$$\mathrm{TC}(T;\mathbb{Z}_p)=\lim_{r\geq 1}\mathrm{TC}(T)/p^r$$

for derived p-complete topological cyclic homology. For $n\geq 0$ define

$$F_p^n\mathrm{TC}(T;\mathbb{Z}_p)=\mathrm{fib}\big(\mathrm{TC}(T;\mathbb{Z}_p)\longrightarrow \mathrm{TC}(T;\mathbb{Z}_p)/p^n\big),$$

where the quotient by $p^0=1$ is zero. Then $F_p^\bullet$ is a complete multiplicative filtration with $F_p^0=\mathrm{TC}(T;\mathbb{Z}_p)$ and

$$\mathrm{gr}^n_{F_p}\mathrm{TC}(T;\mathbb{Z}_p)\simeq \mathrm{TC}(T;\mathbb{Z}_p)/p.$$

For every small canonical family, the natural map

$$F_p^\bullet\mathrm{TC}(A;\mathbb{Z}_p)\longrightarrow \operatorname*{Tot}_q\prod_{\alpha\in I^{q+1}}F_p^\bullet\mathrm{TC}(B_\alpha;\mathbb{Z}_p)$$

is an equivalence in $\mathrm{Fil}^\wedge(\mathrm{Sp})$*, compatibly with multiplication.*

Proof. For every r, the functor $X\mapsto X/p^r$ is tensoring with the finite, hence dualizable, Moore spectrum $\mathbb{S}/p^r$; it therefore preserves all limits. Derived p-completion is the inverse limit of these limit-preserving functors, so Fubini for limits shows that it too preserves all limits. For spectra this inverse-limit functor is $\mathbb{S}/p$-localization [25, Corollary 5.14(e)]; equivalently, for every (not necessarily bounded below) spectrum X one has

$$X_p^\wedge\simeq \mathrm{map}_{\mathrm{Sp}}(\Sigma^{-1}\mathbb{S}/p^\infty,X)\simeq\lim_{r\geq 1}X/p^r,$$

where $\mathbb{S}/p^\infty=\mathrm{colim}_{r\geq 1}\mathbb{S}/p^r$. In particular it is idempotent, so $X_p^\wedge$ is derived p-complete without a connectivity hypothesis. Applying it to Theorem 4.8 gives

$$\mathrm{TC}(A;\mathbb{Z}_p)\simeq\operatorname*{Tot}_q\prod_\alpha\mathrm{TC}(B_\alpha;\mathbb{Z}_p).$$

The functor F_p^n is formed from a quotient and a fiber and hence preserves this limit for each n. The resulting levelwise comparisons assemble to the asserted filtered equivalence.

If X is derived p-complete, taking the limit of the fiber sequences

$$F_p^nX\longrightarrow X\longrightarrow X/p^n$$

gives

$$\lim_n F_p^nX\simeq\mathrm{fib}\big(X\longrightarrow\lim_n X/p^n\big)\simeq 0.$$

Thus the filtration is complete. Under the canonical identification of F_p^nX with the source of $p^n:X\to X$, the transition $F_p^{n+1}X\to F_p^nX$ is multiplication by p; its cofiber is X/p.

It remains to make multiplicativity precise. The $\mathbb{S}/p$-equivalences are stable under tensoring with an arbitrary spectrum, so p-completion is a symmetric-monoidal localization; as an endofunctor of spectra it is lax symmetric monoidal. Hence the E_∞-multiplication on $\mathrm{TC}(T)$ induces one on $X=\mathrm{TC}(T;\mathbb{Z}_p)$. Identify every F_p^nX with X as above and let

$$F_p^iX\otimes F_p^jX=X\otimes X\xrightarrow{\mu}X=F_p^{i+j}X.$$

These maps are compatible with the transition maps because $\mu(px,y) = p\mu(x,y) = \mu(x,py)$; all higher coherences are inherited from the E_∞-multiplication and the central scalar p. Thus they define a commutative algebra object for the Day-convolution monoidal structure on filtered spectra. The construction is natural in T, so the levelwise descent equivalence above is multiplicative. □

Theorem 5.2 (Filtered canonical descent)**.** *For every small canonical family of connective E_∞-rings:*

1. *the complete Hodge filtration on E_∞-infinitesimal cohomology satisfies family Čech descent;*
2. *the complete circle-equivariant HKR filtration on* THH *satisfies family Čech descent as a multiplicative filtered object;*
3. *for every finite $C_m \subset \mathbb{T}$, termwise C_m-Tate of the HKR tower is complete and satisfies filtered descent, with graded pieces obtained by applying $(-)^{tC_m}$ to the HKR graded pieces;*
4. *for every prime p, derived p-complete* TC *has a complete multiplicative p-adic filtration satisfying family Čech descent;*
5. *after evaluating the HKR and finite-group Tate towers in filtration degree zero, the resulting underlying and C_p-Tate comparisons assemble to descent in* CycSp *and hence to integral* TC*-descent.*

No finite-family, Noetherian, Tor-amplitude, characteristic, or quasisyntomic hypothesis is imposed.

Proof. The five assertions are respectively Corollary 3.3, Theorem 4.3, Corollary 4.6, and Theorems 5.1 and 4.8. □

5.2 The HRW even filtration and its canonical-descent obstruction

Fix a prime p. If X is a connective p-complete cyclotomic E_∞-ring, Hahn–Raksit–Wilson define

$$\mathrm{fil}^n_{\mathrm{ev},p,\mathrm{TC}} X = \lim_{X\to B} \mathrm{fib}\left(\tau_{\geq 2n}(B^{h\mathbb{T}})^\wedge_p \xrightarrow{\varphi^{h\mathbb{T}}-\mathrm{can}} \tau_{\geq 2n}(B^{t\mathbb{T}})^\wedge_p\right). \tag{5.1}$$

Here $n \in \mathbb{Z}$. The limit ranges over maps to even, connective, p-complete cyclotomic E_∞-rings with bounded p-power torsion [10, Definition 1.2.3]. This is a right Kan extension, not the HKR filtration of Section 3. Applying it to $\mathrm{THH}(R)^\wedge_p$ defines a $\mathbb{Z}$-indexed presheaf

$$\mathcal{F}^\bullet_{\mathrm{ev},p}(R) := \mathrm{fil}^\bullet_{\mathrm{ev},p,\mathrm{TC}}(\mathrm{THH}(R)^\wedge_p) \quad \text{in } \mathrm{Fil}_\mathbb{Z}(\mathrm{Sp})$$

on the full category of connective E_∞-rings. Its normalized *ambient even-*TC *graded piece* is

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(R) := \mathrm{gr}^i\, \mathcal{F}^\bullet_{\mathrm{ev},p}(R)[-2i]. \tag{5.2}$$

For a chromatically quasisyntomic connective E_∞-ring R, HRW put

$$\mathrm{fil}^\bullet_{\mathrm{mot}}\mathrm{TC}(R;\mathbb{Z}_p) := \mathrm{fil}^\bullet_{\mathrm{ev},p,\mathrm{TC}}(\mathrm{THH}(R)^\wedge_p)$$

and call its graded pieces the syntomic cohomology of R [10, Definitions 1.3.1 and 1.3.4]. We use the Bhatt–Morrow–Scholze (BMS) normalization

$$\mathbb{Z}_p^{\mathrm{HRW}}(i)(R) := \mathrm{gr}^i_{\mathrm{mot}}\,\mathrm{TC}(R;\mathbb{Z}_p)[-2i]. \tag{5.3}$$

Here p-completion is formed in cyclotomic E_∞-rings. On this subcategory there is therefore a natural identification

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(R) \simeq \mathbb{Z}_p^{\mathrm{HRW}}(i)(R). \tag{5.4}$$

Neither the ambient definition nor the coconnective, negative-weight, and completeness statements below use a chromatically quasisyntomic hypothesis. That hypothesis enters only when we evaluate at the two endpoints: it supplies the identification with HRW syntomic cohomology and the exhaustivity statement used in the endpoint argument.

On an even test object B, both $(B^{h\mathbb{T}})_p^\wedge$ and $(B^{t\mathbb{T}})_p^\wedge$ are even [10, Lemma 2.2.12], so exactness gives

$$\begin{aligned}\mathrm{gr}^i \, \mathrm{fil}^\bullet_{\mathrm{ev},p,\mathrm{TC}} B &\simeq \mathrm{fib}\Big(\Sigma^{2i} H\pi_{2i}\big((B^{h\mathbb{T}})_p^\wedge\big) \longrightarrow \Sigma^{2i} H\pi_{2i}\big((B^{t\mathbb{T}})_p^\wedge\big)\Big) \\ &\simeq \tau_{[2i-1,2i]}\big(\mathrm{TC}(B)_p^\wedge\big).\end{aligned} \tag{5.5}$$

The first line of (5.5) is naturally a fiber in $\mathrm{Mod}_{H\mathbb{Z}}$. This construction is functorial in the even test object, and the forgetful functor $\mathrm{Mod}_{H\mathbb{Z}} \to \mathrm{Sp}$ preserves all limits. Therefore, after the normalizing shift, the right Kan extension gives a canonical lift

$$\mathbb{Z}_p^{\mathrm{ev}}(i) : \mathrm{CAlg}^{\mathrm{cn}} \longrightarrow \mathrm{Mod}_{H\mathbb{Z}} \tag{5.6}$$

for every $i \in \mathbb{Z}$. Each graded-piece functor on $\mathrm{Fil}_{\mathbb{Z}}(\mathrm{Sp})$ preserves all small limits: in a stable category its defining cofiber is a shifted fiber. It therefore commutes with the right Kan limit in (5.1). After the normalizing shift $[-2i]$, every term of (5.5) lies in $\mathrm{Sp}_{[-1,0]}$; since $\mathrm{Sp}_{\le 0}$ is closed under limits, $\mathbb{Z}_p^{\mathrm{ev}}(i)$ takes values in $\mathrm{Sp}_{\le 0}$ uniformly on the entire ambient category.

Every test object B in (5.1) is connective and p-complete. Hence $\mathrm{TC}(B)$ is (-1)-connective [9, Lemma 2.1.24]. If $i < 0$, the truncation window in (5.5) is therefore zero. Commuting the graded-piece functor with the right Kan limit gives the ambient vanishing

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(R) = 0 \qquad (R \in \mathrm{CAlg}^{\mathrm{cn}}, \ i < 0). \tag{5.7}$$

Under (5.4), this also gives $\mathbb{Z}_p^{\mathrm{HRW}}(i)(R) = 0$ for every chromatically quasisyntomic R and every $i < 0$.

Proposition 5.3 (Canonical descent forces Postnikov invariance)**.** *Fix $i \in \mathbb{Z}$. If $\mathbb{Z}_p^{\mathrm{ev}}(i)$ is a sheaf for the canonical topology, then it is a canonical hypersheaf and, for every $A \in \mathrm{CAlg}^{\mathrm{cn}}$, the map*

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(A) \longrightarrow \mathbb{Z}_p^{\mathrm{ev}}(i)(H\pi_0 A) \tag{5.8}$$

is an equivalence.

Proof. Equation (5.5) places the presheaf uniformly in $\mathrm{Sp}_{\le 0}$. Under the sheaf hypothesis, it is therefore a coconnective object of the stable category of sheaves for the canonical topology. Lemma 2.6(2) first upgrades it to a hypersheaf and then applies the constant $H\pi_0 A$ hypercover to obtain (5.8). □

Corollary 5.4 (Canonical graded descent forces endpoint agreement)**.** *If every normalized ambient graded piece $\mathbb{Z}_p^{\mathrm{ev}}(i)$ is a sheaf for the canonical topology, then*

$$\mathrm{TC}(\mathbb{S})_p^\wedge \longrightarrow \mathrm{TC}(H\mathbb{Z})_p^\wedge$$

is an equivalence.

Proof. Proposition 5.3, together with (5.4) for the chromatically quasisyntomic rings $\mathbb{S}$ and $H\mathbb{Z}$, makes every map on associated graded pieces between their even filtrations an equivalence. Completeness follows directly from [10, Definition 1.2.3 and Construction 2.1.17] by the limit argument of [10, Remark 2.1.5]: double-speed Postnikov filtrations are complete, and fibers and small limits preserve completeness. At these two chromatically quasisyntomic endpoints, exhaustivity is [10, Theorem 1.3.5]. Restrict the filtrations to indices $n \geq 0$. The restrictions are complete, and Lemma 3.1 makes the restricted filtered map an equivalence, in particular at stage zero. By (5.7), all transitions in negative weights are equivalences. Exhaustivity therefore identifies stage zero with the realization at both endpoints; HRW's realization theorem identifies those realizations with the displayed p-complete TC spectra. □

The obstruction is detected by a chromatic calculation of the endpoint map for the ambient presheaf (5.2).

Theorem 5.5 (Failure of canonical descent for ambient even-TC graded pieces)**.** *For every prime p, there are infinitely many integers $i > 0$ for which the natural map*

$$\mathbb{Z}_p^{\mathrm{ev}}(i)(\mathbb{S}) \longrightarrow \mathbb{Z}_p^{\mathrm{ev}}(i)(H\mathbb{Z}) \tag{5.9}$$

is not an equivalence. Since both endpoints are chromatically quasisyntomic, this is equivalently a failure of the corresponding map of HRW syntomic complexes under (5.4)*. For every such i, the ambient presheaf $\mathbb{Z}_p^{\mathrm{ev}}(i)$ is not a sheaf for the canonical topology, and hence is not a canonical hypersheaf. Consequently the nonnegative restriction of $\mathcal{F}^{\bullet}_{\mathrm{ev},p}$ is not a* $\mathrm{Fil}(\mathrm{Sp})$*-valued sheaf for the canonical topology, and the full filtration is not a* $\mathrm{Fil}_{\mathbb{Z}}(\mathrm{Sp})$*-valued sheaf for that topology.*

Proof. Both $\mathbb{S}$ and $H\mathbb{Z}$ are chromatically quasisyntomic [10, Examples 1.3.2–1.3.3]. As recalled above, their HRW filtrations are complete and exhaustive. Let $C^{\bullet}$ be the fiber of the resulting filtered map

$$\mathrm{fil}^{\bullet}_{\mathrm{mot}}\mathrm{TC}(\mathbb{S}; \mathbb{Z}_p) \longrightarrow \mathrm{fil}^{\bullet}_{\mathrm{mot}}\mathrm{TC}(H\mathbb{Z}; \mathbb{Z}_p).$$

It is again complete, since inverse limits commute with fibers. It is exhaustive because filtered colimits are exact in spectra and hence commute with the fiber of the two realization maps. Its weight-i graded piece is the fiber of (5.9), shifted by $2i$. Write $D_i := \mathrm{gr}^i C^{\bullet}$. By (5.7), all negative D_i vanish, so every transition $C^{i+1} \to C^i$ with $i < 0$ is an equivalence. Exhaustivity therefore identifies the realization with stage zero, and HRW's realization theorem gives

$$C^0 \simeq \mathrm{fib}(\mathrm{TC}(\mathbb{S})^{\wedge}_p \longrightarrow \mathrm{TC}(H\mathbb{Z})^{\wedge}_p). \tag{5.10}$$

We next show, at every prime, that the spectrum in (5.10) is not $K(2)$-acyclic. Here and below $K(2)$ denotes Morava K-theory at the fixed prime p. The comparison between the classical p-typical construction and modern topological cyclic homology applies to the bounded-below cyclotomic sphere [20, Theorem II.3.8(ii) and the discussion preceding Proposition IV.3.4]. Under this comparison, Rognes's spectrum-level formulation of the Bökstedt–Hsiang–Madsen calculation gives

$$\mathrm{TC}(\mathbb{S})^{\wedge}_p \simeq \mathbb{S}^{\wedge}_p \vee \Sigma(\mathbb{CP}^{\infty}_{-1})^{\wedge}_p \tag{5.11}$$

[24, Section 1.10, Corollary 1.18, Theorem 1.20, and Corollary 1.21]; see also [7, Lemma 5.15, Theorem 5.17, Proposition 5.20, and formula (9.1)]. Therefore

$$K(2) \otimes \mathrm{TC}(\mathbb{S})^{\wedge}_p \neq 0, \tag{5.12}$$

because the sphere summand gives $K(2) \otimes \mathbb{S}_p^\wedge \simeq K(2) \neq 0$.

The target is $K(2)$-acyclic at every prime. At the fixed prime p, the multiplicative cyclotomic trace of Blumberg–Gepner–Tabuada is a unital map of E_∞-rings

$$\mathrm{trc}_{\mathbb{Z}} \colon K^{\mathrm{cn}}(\mathbb{Z}) \longrightarrow \mathrm{TC}_{\mathrm{cl}}(H\mathbb{Z}; p),$$

where the target denotes classical p-typical topological cyclic homology [6, Theorem 1.12]. Since $\mathbb{Z}$ is regular Noetherian, connective and nonconnective algebraic K-theory agree; below we write this spectrum as $K(\mathbb{Z})$. Derived p-completion is lax symmetric monoidal, so the completed target remains a $K(\mathbb{Z})$-module. The comparison

$$\mathrm{TC}_{\mathrm{cl}}(H\mathbb{Z}; p)_p^\wedge \simeq \mathrm{TC}(H\mathbb{Z})_p^\wedge$$

of [20, Theorem II.3.8(ii) and the discussion preceding Proposition IV.3.4] transfers this module structure to the underlying modern TC spectrum. Mitchell proves that $K(n)_* K(\mathbb{Z}) = 0$ for every prime and every $n \geq 2$ [19, Theorem A]. If M is a $K(\mathbb{Z})$-module, the composite

$$M \longrightarrow K(\mathbb{Z}) \otimes M \longrightarrow M$$

of the unit and action maps is the identity. Thus $K(2)\otimes M$ is a retract of $K(2)\otimes K(\mathbb{Z})\otimes M \simeq 0$. Applying this to $M = \mathrm{TC}(H\mathbb{Z})_p^\wedge$ gives

$$K(2) \otimes \mathrm{TC}(H\mathbb{Z})_p^\wedge = 0. \tag{5.13}$$

In particular, (5.13) supplies the required two-primary calculation without a two-primary splitting of $\mathrm{TC}(H\mathbb{Z})_2^\wedge$.

Combining (5.12) with (5.13), and using exactness of smash product, gives

$$K(2) \otimes C^0 \simeq K(2) \otimes \mathrm{TC}(\mathbb{S})_p^\wedge \neq 0. \tag{5.14}$$

Suppose that only finitely many positive graded maps (5.9) failed to be equivalences. Choose N such that $D_i = 0$ for $i > N$. The maps $C^{i+1} \to C^i$ are then equivalences for $i > N$; completeness implies $C^{N+1} = 0$. The cofiber sequences

$$C^{i+1} \longrightarrow C^i \longrightarrow D_i \qquad (0 \leq i \leq N)$$

express C^0 as a finite extension of $D_0, \ldots, D_N$. Ravenel's Bousfield-class calculation gives $K(2) \otimes H\mathbb{F}_p \simeq 0$ [23, Theorem 2.1(i)]. Since p acts trivially on $K(2)$, smashing the cofiber sequence $H\mathbb{Z}_{(p)} \xrightarrow{p} H\mathbb{Z}_{(p)} \to H\mathbb{F}_p$ shows that $K(2) \otimes H\mathbb{Z}_{(p)} \simeq 0$. Since $K(2)$ is p-local, it follows that $K(2) \otimes H\mathbb{Z} \simeq 0$. By (5.6), every D_i is an $H\mathbb{Z}$-module, so its unit and action maps exhibit $K(2) \otimes D_i$ as a retract of

$$K(2) \otimes H\mathbb{Z} \otimes D_i \simeq 0.$$

Thus every D_i is $K(2)$-acyclic. The finite sequence of cofiber sequences displayed above then makes C^0 $K(2)$-acyclic. This contradicts (5.14). Infinitely many positive maps (5.9) therefore fail.

Fix one of the weights just obtained. If $\mathbb{Z}_p^{\mathrm{ev}}(i)$ were a sheaf for the canonical topology, its global coconnective bound and Lemma 2.6(2) would give precisely the forbidden equivalence (5.8). Thus it is not a sheaf for the canonical topology. A canonical hypersheaf is in particular such a sheaf, so hyperdescent fails as well. If the nonnegative restriction of $\mathcal{F}^\bullet_{\mathrm{ev},p}$ were a $\mathrm{Fil}(\mathrm{Sp})$-valued sheaf for the canonical topology, then each of its nonnegative graded pieces would be a sheaf for that topology, because every graded-piece functor preserves all limits. This contradicts the failure in infinitely many positive weights. Restriction along $\mathbb{N}^{\mathrm{op}} \hookrightarrow \mathbb{Z}^{\mathrm{op}}$ also preserves limits, so the full $\mathbb{Z}$-indexed filtration cannot be a $\mathrm{Fil}_{\mathbb{Z}}(\mathrm{Sp})$-valued sheaf for the canonical topology. □

Remark 5.6 (Exact scope of the obstruction). The theorem does not identify which positive weights fail and makes no assertion about weight zero. For $i < 0$, (5.7) says that the ambient presheaf is identically zero; in particular, it is a sheaf and a hypersheaf for the canonical topology. The theorem also does not contradict HRW's proved descent theorem for p-completely eff maps [10, Corollary 2.2.17(3)], the comparison with the BMS filtration on p-complete quasisyntomic discrete rings [10, Theorem 1.2.4], or the smooth characteristic-p comparison of Antieau–Nikolaus [2, Theorem 6.23].

The constant Postnikov hypercover is not the Čech nerve of $\mathbb{S} \to H\mathbb{Z}$: the latter has terms $H\mathbb{Z}^{\otimes_{\mathbb{S}}(q+1)}$, rather than constant value $H\mathbb{Z}$. Thus endpoint disagreement first detects failure of hyperdescent. The stronger failure of descent for the canonical topology follows from (5.5): the normalized ambient graded piece has a global coconnective bound, so Lemma 2.6 would make any sheaf for the canonical topology hypercomplete.

6 Geometric descent and four-vertex squares

We now globalize affine canonical descent. We first pass to quasi-compact and quasi-separated (qcqs) spectral schemes, then lift finitely presented classical h-covers to canonical spectral covers. This yields K-theoretic cdh descent through K^{inv}. The final proposition separates full Čech descent from cartesianness of a fixed four-vertex square. Schemes in this section are connective and qcqs, and the covering morphisms are qcqs. For a localizing invariant E, we write $E(X) = E(\mathrm{Perf}(X))$; K denotes nonconnective algebraic K-theory.

6.1 Canonical descent on spectral schemes

Theorem 6.1 (Canonical descent on spectral schemes)**.** *Cyclotomic* THH *and integral* TC *satisfy Čech descent for every small qcqs covering family in the canonical Grothendieck topology on connective qcqs spectral schemes.*

Proof. The assertion is local on the target for the Zariski topology, so we may assume that the target X is affine. Refine each qcqs source Y_i by a finite affine Zariski cover $\{V_{ij} \to Y_i\}_j$. The composite family $\{V_{ij} \to X\}_{i,j}$ is canonical by base change, composition, and refinement in the canonical topology. Proposition 2.2 and Theorem 4.8 therefore give descent for this affine family.

Compare the two covers by their bisimplicial Čech object. In one simplicial direction it is the Čech object of $\{Y_i \to X\}_i$; over each of its terms, the other direction is the finite Zariski refinement induced by the V_{ij}. Cyclotomic THH and TC satisfy finite Zariski descent on qcqs spectral schemes by localization for perfect complexes [5, 13]. Iterated totalization thus identifies descent for the refined family with descent for the original one. Applying the same comparison to a finite affine cover of a general qcqs target proves the theorem. Smallness is used only to remain in the fixed universe; no finiteness condition is imposed on the canonical family. □

This is Čech, or sheaf, descent. It does not assert hyperdescent: a spectrum-valued sheaf need not be hypercomplete [8, Example 4.15]. This is the convention fixed after Proposition 2.2; compare the HRW canonical-descent obstruction in Theorem 5.5.

6.2 From h-covers to algebraic K-theory

Proposition 6.2 (Lifting a classical h-cover)**.** *Let $\{A \to B_i\}_{i=1}^{r}$ be a finite family of connective spectral rings. If $\{\mathrm{Spec}\, \pi_0 B_i \to \mathrm{Spec}\, \pi_0 A\}_{i=1}^{r}$ is a finitely presented classical h-cover, then the spectral family is canonical.*

Proof. Put $B = \prod_{i=1}^{r} B_i$, corresponding to the disjoint union of the sources. The entire finite product records the covering family; its individual members need not be surjective. In the noetherian case, Bhatt–Scholze prove that the structure algebra of a finitely presented h-cover is descendable [4, Proposition 11.25]; descendability gives universally convergent Amitsur descent [18, Proposition 3.22]. Since $\operatorname{Spec}\pi_0 B \to \operatorname{Spec}\pi_0 A$ is of finite presentation, the approximation lemma for affine v-covers [4, Lemma 2.12] descends both the map and the covering property to a common stage. Thus there is an h-cover $A_0 \to B_0$ between rings finitely presented over $\mathbb{Z}$ such that the original map is its base change:

$$\pi_0 B \cong \pi_0 A \otimes_{A_0} B_0.$$

The noetherian case and spectral base change therefore make

$$H\pi_0 A \longrightarrow B' = H\pi_0 A \otimes_{HA_0} HB_0$$

canonical. Since $\pi_0 B' \cong \pi_0 B$, the truncation map $B' \to H\pi_0 B$ has 1-connective fiber and is canonical by Lemma 2.4. Thus $H\pi_0 A \to H\pi_0 B$ is canonical.

The truncation map $A \to H\pi_0 A$ is canonical for the same reason. Its composite with $H\pi_0 A \to H\pi_0 B$ factors as

$$A \longrightarrow B \longrightarrow H\pi_0 B.$$

Right cancellation in Lemma 2.3 shows that $A \to B$ is canonical. Finally, base change of a finite product decomposition preserves its orthogonal idempotents. Hence, for every $q \geq 0$,

$$B^{\otimes_A (q+1)} \simeq \prod_{(i_0,\dots,i_q)\in\{1,\dots,r\}^{q+1}} B_{i_0} \otimes_A \cdots \otimes_A B_{i_q}.$$

Thus the Amitsur object of $A \to B$ is exactly the family Čech object, which proves the assertion. □

Finite affine refinements and finite Zariski descent globalize Proposition 6.2; the refinement comparison is the bisimplicial Čech comparison used in the proof of Theorem 6.1.

Definition 6.3 (Spectral cdh cover)**.** A finite spectral cdh covering family is a finite family of morphisms that are locally almost of finite presentation and whose classical truncation is a finitely presented classical cdh covering family.

Proposition 6.4 (Geometric h- and cdh descent)**.** *Let $\{Y_i \to X\}_{i=1}^{r}$ be a finite family of morphisms locally almost of finite presentation between connective qcqs spectral schemes. If its classical truncation is a finitely presented h-cover, then* THH *and* TC *satisfy Čech descent for the family. If the family is spectral cdh, then nonconnective K-theory satisfies the full derived Čech formula*

$$K(X) \simeq \operatorname*{Tot}_q K(Y^{[q]}), \qquad Y^{[q]} = \coprod_{(i_0,\dots,i_q)} Y_{i_0} \times_X \cdots \times_X Y_{i_q}. \tag{6.1}$$

Proof. After passing to a finite affine cover of X and finite affine refinements of its inverse images, Proposition 6.2 shows that the refined family is canonical. Its Čech formula follows from Theorem 6.1, and finite Zariski descent together with the bisimplicial refinement comparison gives the formula for the original family. This proves the THH and TC assertions. For K let

$$K^{\mathrm{inv}} = \mathrm{fib}(K \longrightarrow \mathrm{TC}).$$

Land–Tamme prove that K^{inv} is a truncating localizing invariant and that every truncating invariant is a cdh sheaf on classical qcqs schemes [13, Corollary 3.6, proof, and Theorem A.2].

Consequently $K^{\mathrm{inv}}(T) \simeq K^{\mathrm{inv}}(t_0 T)$ on affines; finite Zariski descent globalizes this equivalence to every qcqs spectral scheme T. Apply it to all the terms $Y^{[q]}$. Classical truncation preserves the finite fiber products of connective spectral schemes, so classical cdh descent gives (6.1) for K^{inv}. It holds for TC by the first part. Totalization is an exact limit of spectra, and the fiber sequence $K^{\mathrm{inv}} \to K \to \mathrm{TC}$ proves the formula for K. □

Formula (6.1) uses every level of the derived Čech nerve. The four vertices of a spectral abstract blow-up square, by contrast, need not form a cartesian K-theory square. The next example makes the distinction concrete.

6.3 The full Čech nerve and four-vertex squares

Proposition 6.5 (A noncartesian four-vertex square)**.** *There is a finite spectral abstract blow-up square, locally almost of finite presentation, whose four K-theory vertices do not form a cartesian square.*

Proof. The square. Let $R = \mathbb{Q}[t]$, let $N = R/(t^2)$, and form the discrete square-zero algebra $A_0 = R \oplus N$. Put $C_0 = R$, write $A = HA_0$ and $C = HC_0$, and let $B = A//t$ denote the free E_∞-quotient obtained by adjoining a nullhomotopy of t; set

$$D = C \otimes_A B.$$

Since t is regular in C, $D \simeq C//t \simeq H\mathbb{Q}$. A direct Koszul calculation gives

$$\pi_0 B \cong \mathbb{Q}[\epsilon]/(\epsilon^2), \qquad \pi_1 B \cong tN \cong \mathbb{Q}, \qquad \pi_i B = 0 \quad (i > 1). \tag{6.2}$$

The quotient $A \to C$ is finite and almost of finite presentation and becomes an equivalence after inverting t, since $N[t^{-1}] = 0$. The map $A \to B$ is a one-cell finitely presented spectral closed immersion supported on $V(t)$. The derived pullback square with fourth corner D is therefore a finite spectral abstract blow-up square.

The derived self-intersection. Its derived self-intersection is not trivial. Indeed,

$$B \otimes_A B \simeq B \otimes_{\mathbb{Q}} \Lambda_{\mathbb{Q}}(\eta), \qquad |\eta| = 1, \tag{6.3}$$

because $\mathbb{Q} \otimes^L_{\mathbb{Q}[t]} \mathbb{Q} \simeq \Lambda_{\mathbb{Q}}(\eta)$.

The K-theoretic obstruction. Suppose the four K-theory vertices were cartesian. The associated Mayer–Vietoris sequence would make

$$K_2(D) \otimes \mathbb{Q} \longrightarrow K_1(A) \otimes \mathbb{Q} \longrightarrow (K_1(B) \oplus K_1(C)) \otimes \mathbb{Q}$$

exact. The subgroup $1 + tN \cong (\mathbb{Q}, +)$ injects into $K_1(A)$: the map from units is split by the determinant. This subgroup maps to the identity in both $K_1(B)$ and $K_1(C)$. On the other hand $K_2(D) \otimes \mathbb{Q} = K_2(\mathbb{Q}) \otimes \mathbb{Q} = 0$, by Quillen localization for $\mathbb{Z} \subset \mathbb{Q}$ [21], since the adjacent groups are torsion. This contradicts exactness.

The higher terms of the full derived Čech nerve contain the self-intersections in (6.3). Thus the counterexample is compatible with Proposition 6.4: it rules out four-vertex excision, not full Čech descent. □

The next section packages the higher Čech terms into a formal defect and asks when completion makes that defect weakly pro-zero.

7 Formal towers and canonical Čech compression

At every level of a formal completion tower, canonical descent gives a full derived Čech resolution. Formal pro-excision asks whether these resolutions weakly compress to their four-vertex pullbacks. This section defines the resulting formal defect and proves its vanishing for finite modifications. Section 8 treats a proper modification over a classical base; Sections 9 and 10 compare the general spectral case with that one.

7.1 Weak pro-equivalences and intrinsic completion

Definition 7.1 (Weak pro-equivalence)**.** A map $P \to Q$ in $\mathrm{Pro}(\mathrm{Sp})$ is a *weak pro-equivalence* if

$$\{\tau_{\le r} P\} \longrightarrow \{\tau_{\le r} Q\}$$

is an equivalence of pro-spectra for every integer r. A square in $\mathrm{Pro}(\mathrm{Sp})$ is *weakly cartesian* if its total fiber is weakly zero.

This notion retains exactly the uniformity needed below: for each fixed output degree there must be a sufficiently deep stage of the completion tower, but that stage may depend on the degree.

Let $X = \operatorname{Spec} A$ and let $Z \subset |X|$ be a closed subset whose open complement is quasi-compact. Choose $f_1, \ldots, f_s \in \pi_0 A$ with $Z = V(f_1, \ldots, f_s)$. We use the free spectral Koszul quotients

$$A_a = A//(f_1^a, \ldots, f_s^a), \qquad a \ge 1. \tag{7.1}$$

Thus A_a is obtained by freely adjoining nullhomotopies of the displayed powers. Equivalently, it is the pushout of connective E_∞-rings

$$A_a = A \otimes_{\mathrm{Sym}_{\mathbb{S}}(\mathbb{S}^{\oplus s})} \mathbb{S},$$

where the two maps send the ith generator to f_i^a and to zero, respectively. This tower is a cofinal affine presentation of the intrinsic formal completion $X_Z^\wedge$ of [17, Section 8.1].

We shall globalize the affine construction by finite limits. The finiteness in the following elementary lemma is important: the diagram is obtained by iterated binary Mayer–Vietoris squares, not by truncating an infinite Čech nerve.

Lemma 7.2 (Finite affine Mayer–Vietoris presentations)**.** *Every qcqs spectral scheme T admits a finite rooted system of binary Zariski covers*

$$W = W' \cup W'', \qquad \begin{array}{ccc} W' \cap W'' & \longrightarrow & W'' \\ \big\downarrow & & \big\downarrow \\ W' & \longrightarrow & W, \end{array}$$

whose terminal members are affine. Consequently, if F is a presheaf valued in an ∞-category with finite limits and F carries every binary Zariski cover to a pullback square, then $F(T)$ is the limit of a fixed finite diagram of its values on affine schemes. The same assertion holds after pullback along a qcqs morphism, after making finitely many further affine refinements of the pulled-back terminal members.

Proof. Choose a finite affine cover $T = U_1 \cup \cdots \cup U_m$. Induction on m reduces the construction to a finite presentation of $(U_1 \cup \cdots \cup U_{m-1}) \cap U_m$. This is a quasi-compact open of the affine scheme U_m, hence is a finite union of principal affine opens. It remains to treat a union $D(g_1)\cup\cdots\cup D(g_t)$ inside one affine scheme. A second induction on t uses the binary cover by $D(g_1)\cup\cdots\cup D(g_{t-1})$ and $D(g_t)$; their intersection is

$$D(g_1g_t)\cup\cdots\cup D(g_{t-1}g_t)\subset D(g_t),$$

so the second induction terminates with principal affine opens. This constructs a finite tree of binary covers. Applying binary Mayer–Vietoris descent at each vertex expresses $F(T)$ by a finite iterated pullback. Pulling back the tree preserves its binary covers; quasi-compactness and quasi-separatedness provide finite affine refinements of its finitely many terminal members. □

For a qcqs spectral scheme we now fix one such finite affine Zariski–Mayer–Vietoris presentation and use the towers (7.1) on its affine members. For every arrow in the finite presentation, the restricted tower and any tower chosen on the target represent the same intrinsic formal completion and hence admit a common cofinal refinement. Since there are only finitely many arrows, all these refinements can be made simultaneously: first replace the indexing set by their finite product and then use the cofinal diagonal $\mathbb{N}\to\mathbb{N}^d$. Thus the entire finite Mayer–Vietoris diagram is represented levelwise over one formal parameter. We write $E(X^\wedge_Z)$ for the resulting pro-spectrum. A second finite choice of generators has a common cofinal refinement with the first and therefore defines the same pro-object. Throughout, formal completion means this intrinsic pro-object. Once a presentation has been chosen, the number of finite-limit operations occurring in it is fixed and independent of the formal parameter.

7.2 The formal defect as a Čech compression fiber

We first give the construction when the chosen tower is represented by finitely presented spectral closed immersions $X_a\hookrightarrow X$ with $|X_a|=Z$. On a general qcqs scheme, the same construction is made on the affine terminal members of the fixed binary Mayer–Vietoris presentation and then assembled by its finite iterated pullbacks; we use the same notation for the resulting finite diagram.

Let $p:Y\to X$ be proper and locally almost of finite presentation, and suppose that it is an equivalence over $U=X\setminus Z$. Write $Y_a=Y\times_X X_a$, and put

$$P_{E,a}=E(Y)\times_{E(Y_a)}E(X_a).$$

For an invariant E, define the formal defect

$$\Phi_E(p)=\{\mathrm{fib}(E(X)\longrightarrow P_{E,a})\}_a. \tag{7.2}$$

Stability identifies this with

$$\mathrm{fib}\left(\{\mathrm{fib}(E(X)\to E(X_a))\}_a\longrightarrow\{\mathrm{fib}(E(Y)\to E(Y_a))\}_a\right),$$

so the formal square is weakly cartesian precisely when $\Phi_E(p)$ is weakly zero.

Canonical descent gives a Čech model for this defect. Put

$$W_a=Y\amalg X_a,\qquad C_a^q=W_a^{\times_X(q+1)}.$$

Thus $C_a^\bullet$ is the full derived Čech nerve of the two-member family $\{Y,X_a\}\to X$. Degree-zero evaluation, together with the degree-one Čech coherence on the mixed component Y_a, defines a natural map

$$\rho_{E,a}:\operatorname*{Tot}_{[q]\in\Delta}E(C_a^q)\longrightarrow P_{E,a}. \tag{7.3}$$

Proposition 7.3 (The formal defect as a Čech compression fiber)**.** *For $E = \mathrm{TC}$ and for nonconnective K-theory, the augmentations induce a natural equivalence of pro-spectra*

$$\operatorname{const} E(X) \simeq \left\{ \operatorname*{Tot}_{[q]\in\Delta} E(C_a^q) \right\}_a . \tag{7.4}$$

Under this equivalence, the pro-fiber of $\{\rho_{E,a}\}_a$ is $\Phi_E(p)$. Consequently, proper formal pro-excision is equivalent to the assertion that the full canonical Čech resolutions weakly compress to the four-vertex pullbacks:

$$\left\{ \operatorname*{Tot}_{[q]\in\Delta} E(C_a^q) \right\}_a \xrightarrow{\sim_{\mathrm{w}}} \{P_{E,a}\}_a .$$

Proof. For every a, the family $\{Y, X_a\} \to X$ is a finite spectral cdh cover: on classical truncations, p is a proper isomorphism over U and X_a is a finitely presented closed thickening with underlying space Z. Proposition 6.4 gives the augmented Čech equivalence for both TC and K. These equivalences are natural in a, which proves (7.4).

Additivity gives $E(C_a^0) = E(Y \amalg X_a) \simeq E(Y) \times E(X_a)$. The totalization maps to this degree-zero term. The degree-one coherence, restricted to either mixed component of C_a^1, identifies the two resulting maps to $E(Y_a)$, so degree-zero evaluation factors canonically through $P_{E,a}$. After the preceding descent equivalence, $\rho_{E,a}$ is the usual map $E(X) \to P_{E,a}$. Its fiber is therefore $\Phi_E(p)_a$ by (7.2). Applying the construction on each affine terminal member and taking the fixed finite iterated Mayer–Vietoris pullbacks proves the qcqs statement. □

7.3 Uniform control of totalization

Canonical descent supplies the complete resolution (7.4); the remaining problem is uniform compression of its higher derived self-intersections. The following example records the failure of pointwise control in each Čech degree.

Example 7.4 (Weak pro-zero diagrams and totalization)**.** Under stable Dold–Kan, let a tower of cosimplicial spectra $\{D_r^\bullet\}_r$ have normalized cochains

$$N^q D_r = \Sigma^q H(\mathbb{Z}/p^{q+1})$$

with zero differential, and let every tower transition act by multiplication by p. For every fixed q, the normalized cochain tower is weakly zero, since the composite of $q+1$ transitions acts by $p^{q+1} = 0$. Nevertheless,

$$\operatorname{Tot} D_r^\bullet \simeq \prod_{q\geq 0} H(\mathbb{Z}/p^{q+1}),$$

and no fixed transition in the tower kills π_0 of this product. Thus degreewise weak pro-zero information does not by itself control an infinite totalization. In particular, a fixed Postnikov truncation does *not* make an arbitrary totalization depend on only finitely many cosimplicial degrees. The formal-functions argument below filters the formal defect itself and thereby supplies the uniformity missing from this example.

7.4 Koszul completion and finite modifications

We shall apply a connectivity form of the Land–Tamme pro-Milnor theorem. A spectrum-valued localizing invariant E is *ℓ-connective* if, for every n-connective π_0-isomorphism $A \to$

B of connective E_1-rings with $n \geq 1$, the map $E(A) \to E(B)$ is $(n+\ell)$-connective [13, Definition 2.5]. Nonconnective K-theory and integral TC are 1-connective, while THH is 0-connective [13, Example 2.6].

The first input says that a formal completion is idempotent under derived self-intersection. This is a statement about the entire pro-system, not about any one thickening.

Lemma 7.5 (Formal self-intersection)**.** *For the tower* (7.1)*, multiplication induces an equivalence*

$$\{A_a \otimes_A A_a\}_a \xrightarrow{\simeq} \{A_a\}_a \qquad \textit{in } \mathrm{Pro}(\mathrm{CAlg}_A^{\mathrm{cn}}).$$

The analogous map is an equivalence for every fixed finite number of tensor factors.

Proof. By [17, Lemma 8.1.2.2 and Proposition 8.1.5.2], the filtered prestack

$$\operatorname*{colim}_a \operatorname{Spec} A_a \longrightarrow \operatorname{Spec} A$$

is the formal-completion subfunctor, hence a monomorphism. Its fiber product with itself over $\operatorname{Spec} A$ is therefore itself. Filtered colimits commute with finite limits in spaces, and the diagonal is cofinal in the independent thickening indices. The full faithfulness of $\mathrm{Ind}(\mathrm{Aff}_A) \to \mathrm{PSh}(\mathrm{Aff}_A)$ then identifies the corresponding ind-affine objects. Passing to opposite rings gives the displayed pro-equivalence. Repeating the same argument proves the assertion for any fixed finite self-intersection. □

Remark 7.6. The qualifier "fixed" is essential. The lemma gives no transition in the formal tower which works simultaneously for self-intersections of every arity, and hence no direct statement about an infinite Čech totalization.

The second input recovers a supported almost-perfect module from the same tower. Its weaker conclusion is the source of the word "weak" in the main formal theorem.

Lemma 7.7 (Completion of supported almost-perfect modules)**.** *Let M be a bounded-below almost-perfect A-module and suppose that $M[f_i^{-1}] \simeq 0$ for $1 \leq i \leq s$. Then*

$$M \longrightarrow \{M \otimes_A A_a\}_a$$

is a weak equivalence of pro-A-modules. After tensoring with any connective A-module, the resulting comparison is still a weak equivalence.

Proof. Put $I = (f_1, \ldots, f_s)$ and fix an output degree r. Choose a lower bound c for M. Because every A_a is connective over A, tensoring with A_a preserves lower connectivity. In particular,

$$\tau_{\leq r}(M \otimes_A A_a) \simeq \tau_{\leq r}(\tau_{[c,r]} M \otimes_A A_a). \tag{7.5}$$

Thus no homotopy group of M above degree r can enter the displayed truncation after base change.

If $r < c$, the required truncations vanish, so assume $r \geq c$ and put $Q = \tau_{[c,r]} M$. Almost perfectness makes $\tau_{\leq r} M$ finitely r-presented and therefore compact in $(\mathrm{Mod}_A)_{\leq r}$ [17, Remarks 2.7.0.2, 2.7.1.3, and 2.7.1.4]. For $N \in (\mathrm{Mod}_A)_{[c,r]}$, orthogonality for the standard t-structure gives

$$\mathrm{Map}_A(Q, N) \simeq \mathrm{Map}_A(\tau_{\leq r} M, N).$$

Consequently Q is compact in the bounded truncated category $(\mathrm{Mod}_A)_{[c,r]}$. For each i, compatibility of the standard t-structure with filtered colimits gives

$$\operatorname{colim}(Q \xrightarrow{f_i} Q \xrightarrow{f_i} \cdots) \simeq \tau_{[c,r]}(M[f_i^{-1}]) \simeq 0.$$

Applying $\mathrm{Map}_A(Q, -)$ and using compactness, the class of id_Q therefore vanishes at a finite stage of this colimit. Equivalently, some power $f_i^{N_i}$ acts nullhomotopically on Q. Since there are only finitely many i, we may increase the exponents simultaneously. The usual pigeonhole argument for $I = (f_1, \ldots, f_s)$ then supplies a single integer e such that

$$I^e \pi_j M = 0 \qquad (c \le j \le r).$$

We first treat one Postnikov layer $N \simeq \Sigma^j H\pi_j M$ annihilated by I^e. Set

$$R_e = H(\pi_0 A / I^e).$$

Put $N_0 = H\pi_j M$, so $N \simeq \Sigma^j N_0$ and N_0 is a connective discrete R_e-module. In the discrete case at the end of the proof of [17, Lemma 8.1.2.3, after the reduction (*)], the nilpotence of the image of I gives an equivalence

$$\{A_a \otimes_A R_e\}_a \simeq \mathrm{const}\, R_e. \tag{7.6}$$

Here the equivalence is in $\mathrm{Pro}(\mathrm{CAlg}^{\mathrm{cn}}_{R_e})$, and in particular is an equivalence of the underlying pro-R_e-modules, rather than only an assertion about their inverse limits. Applying the levelwise functor $C \mapsto C \otimes_{R_e} N_0$ to (7.6) gives an equivalence in $\mathrm{Pro}(\mathrm{Mod}_{R_e})$:

$$\{A_a \otimes_A N_0\}_a \simeq \{(A_a \otimes_A R_e) \otimes_{R_e} N_0\}_a \simeq \mathrm{const}\, N_0.$$

Suspending this equivalence by j gives the corresponding equivalence for the original layer N. The object $\tau_{[c,r]} M$ has only finitely many Postnikov layers, all annihilated by the same I^e. Induction through its finite Postnikov fiber sequences therefore gives

$$\{\tau_{\le r}(\tau_{[c,r]} M \otimes_A A_a)\}_a \simeq \mathrm{const}\, \tau_{[c,r]} M.$$

Together with (7.5), this proves the asserted weak equivalence in degree r. The exponent e may depend on r, which accounts for the weak, rather than strict, conclusion.

Finally, after suspending by $-c$ the comparison is a weak equivalence between connective pro-modules. Hence [13, Lemma 2.29] shows that tensoring it with any connective A-module preserves weak equivalence; suspending back proves the last assertion. $\square$

We can now treat finite modifications without passing through classical geometry.

Theorem 7.8 (Finite spectral formal pro-excision)**.** *Let $p : Y \to X$ be a finite morphism of connective qcqs spectral schemes which is locally almost of finite presentation. Let $Z \subset |X|$ be a closed subset whose open complement is quasi-compact, and assume that p is an equivalence over $X \setminus Z$. For every ℓ-connective localizing invariant E, the square*

$$\begin{tikzcd} E(X) \arrow[r] \arrow[d] & E(X^\wedge_Z) \arrow[d] \\ E(Y) \arrow[r] & E(Y^\wedge_{p^{-1}Z}) \end{tikzcd}$$

is weakly cartesian in $\mathrm{Pro}(\mathrm{Sp})$.

Proof. First suppose that $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$. Put $J = \mathrm{fib}(A \to B)$ and let $B_a = B \otimes_A A_a$. Finiteness and the property of being almost of finite presentation imply that B, and hence J, is bounded below and almost perfect as an A-module [17, Corollary 5.2.2.2]. Since p is an equivalence away from Z, one has $J[f_i^{-1}] \simeq 0$ for every chosen generator f_i.

Lemma 7.7 identifies J weakly with $\{J \otimes_A A_a\}_a$. Consequently the pro-square of underlying modules associated with

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A_a & \longrightarrow & B_a \end{array}$$

is weakly cartesian. By the definition of B_a, the canonical base-change map $A_a \otimes_A B \to B_a$ is an equivalence at every stage. Fix an integer $n \geq 1$. The second hypothesis of the Land–Tamme pro-Milnor theorem is therefore weakly n-connective, while its first hypothesis is the weak cartesianness just proved. The proof of [13, Theorem 2.32], with [13, Corollary 2.31] in place of its K-theoretic connectivity estimate as in [13, Remark 2.36], gives a weakly $(n + \ell - 1)$-cartesian square. Since the base-change map is an equivalence, the argument applies for every n; hence the square of E-theory pro-spectra is weakly cartesian.

For general X, apply the affine argument to the fixed finite binary presentation of Lemma 7.2; because p is finite, the inverse image of every affine member is affine. Localization for perfect complexes gives binary Zariski Mayer–Vietoris descent for every localizing invariant, as recalled in the proof of Theorem 6.1. Taking the resulting finite iterated Mayer–Vietoris pullback therefore globalizes the affine comparison. Its finite length is independent of a, so it preserves weak equivalences after every fixed Postnikov truncation. □

For $E = K$ or TC, Proposition 7.3 reformulates the theorem by saying that the full canonical Čech resolution weakly compresses along the formal tower. The supported module J and the pro-Milnor theorem control the formal defect at once.

Remark 7.9 (Choice of completion tower). The free spectral quotient $A//f^a$ is a cell attachment in $\mathrm{CAlg}^{\mathrm{cn}}$; its underlying A-module is not, in general, the two-term cofiber of multiplication by f^a. Accordingly, the proof uses Lemma 7.7 rather than the formal cofiber sequence $M \xrightarrow{f^a} M \to M \otimes_A (A//f^a)$, which belongs to a different Koszul model. Section 8 compares the free spectral and animated towers without identifying them term by term.

8 The comparison over a classical base

The finite theorem is entirely spectral. To pass from it to a proper modification over a classical base, we compare the free spectral completion tower with the animated Koszul tower used in derived pro-cdh descent. The comparison is weak, but its connectivity is uniform and it survives the scalar extensions needed on affine charts.

Let R be a discrete ring and let $I = (f_1, \ldots, f_s)$. Write

$$K_a^{\mathrm{sp}} = HR//_{E_\infty}(f_1^a, \ldots, f_s^a)$$

for the free spectral quotient and

$$K_a^{\mathrm{an}} = R//_{\mathrm{an}}(f_1^a, \ldots, f_s^a)$$

for the animated Koszul quotient. If U_{sp} denotes the underlying connective E_∞-ring functor, the comparison between the two universal constructions gives a map $K_a^{\mathrm{sp}} \to U_{\mathrm{sp}}(K_a^{\mathrm{an}})$.

8.1 The completion-tower comparison

Proposition 8.1 (Spectral–animated completion comparison). *The map*

$$\{K_a^{\mathrm{sp}}\}_a \longrightarrow \{U_{\mathrm{sp}}(K_a^{\mathrm{an}})\}_a$$

is a weak equivalence of connective pro-E_1-rings. It remains a weak equivalence after arbitrary connective central scalar extension. Every ℓ-connective localizing invariant, including K, THH, *and* TC, *sends it to a weak equivalence of pro-spectra.*

In this statement, a weak equivalence of pro-E_1-rings means a map whose underlying map of pro-spectra is a weak equivalence in the sense of Definition 7.1.

A central scalar extension means tensoring over HR with a connective E_1-HR-algebra for which the structural HR-action is central. This form includes the connective commutative algebras that occur on the affine charts below.

Proof. Pro-idempotence. We first record pro-idempotence on both sides. For the spectral tower, Lemma 7.5 gives

$$\{K_a^{\mathrm{sp}} \otimes_{HR} K_a^{\mathrm{sp}}\}_a \simeq \{K_a^{\mathrm{sp}}\}_a. \tag{8.1}$$

The universal property and affine Koszul presentation of animated completion in [12, Section 2.5] give the parallel equivalence

$$\{K_a^{\mathrm{an}} \otimes_R^{\Delta} K_a^{\mathrm{an}}\}_a \simeq \{K_a^{\mathrm{an}}\}_a \qquad \text{in } \mathrm{Pro}(\mathrm{CAlg}_R^{\Delta}). \tag{8.2}$$

Indeed, the animated completion is again a monomorphism of prestacks; filtered colimits commute with finite limits, and the diagonal is cofinal in the independent thickening indices. The functor $U_{\mathrm{sp}} : \mathrm{CAlg}^{\Delta} \to \mathrm{CAlg}^{\mathrm{cn}}$ preserves small limits and colimits and is conservative [12, Section 2.1]. Hence (8.2) remains an actual pro-equivalence after passing to connective spectral rings.

Uniform connectivity. Next we obtain a uniform connectivity bound for $K_a^{\mathrm{sp}} \to U_{\mathrm{sp}}(K_a^{\mathrm{an}})$. Put

$$P = \mathrm{Sym}_{HR}(HR^{\oplus s}), \qquad Q = H(R[t_1, \ldots, t_s]).$$

The canonical map $P \to Q$ identifies $\pi_0 P \cong R[t_1, \ldots, t_s]$ with $\pi_0 Q$; since both rings are connective, it is 1-connective. Regard $M = HR$ as a Q-algebra by $t_i \mapsto f_i^a$, and regard $N = HR$ as a Q-algebra by $t_i \mapsto 0$. Then

$$K_a^{\mathrm{sp}} \simeq M \otimes_Q (Q \otimes_P N), \qquad U_{\mathrm{sp}}(K_a^{\mathrm{an}}) \simeq M \otimes_Q N.$$

The section

$$N \longrightarrow Q \otimes_P N$$

of the augmentation $Q \otimes_P N \to N$ is obtained from $P \to Q$ by base change along $P \to N$, and is therefore 1-connective. Since it is a section, the fiber of the augmentation is the cofiber of this section and is therefore 2-connective. Base change along the connective Q-algebra M preserves this lower bound. Thus every comparison $K_a^{\mathrm{sp}} \to U_{\mathrm{sp}}(K_a^{\mathrm{an}})$ is uniformly 2-connective.

The fiber tower. Let J_a be the fiber of this comparison. For a map of associative rings write

$$L_{B/A}^{\mathrm{ass}} = \mathrm{fib}(B \otimes_A B \longrightarrow B).$$

For brevity put $A_a = K_a^{\mathrm{sp}}$ and $B_a = U_{\mathrm{sp}}(K_a^{\mathrm{an}})$. Regard $L_{A_a/HR}^{\mathrm{ass}}$ as an A_a-bimodule and $L_{B_a/HR}^{\mathrm{ass}}$ as a B_a-bimodule. The self-intersection equivalences (8.1) and (8.2) say that both of these systems are weakly zero as pro-bimodules. Moreover, they are connective: multiplication admits a section, so its fiber is a retract of a connective bimodule.

The transitivity triangle is

$$B_a \otimes_{A_a} L_{A_a/HR}^{\mathrm{ass}} \otimes_{A_a} B_a \longrightarrow L_{B_a/HR}^{\mathrm{ass}} \longrightarrow L_{B_a/A_a}^{\mathrm{ass}}.$$

Apply [13, Lemma 2.29] first to extend the weakly zero right-A_a-module $L^{\mathrm{ass}}_{A_a/HR}$ along $A_a \to B_a$, and then again to extend the resulting weakly zero left-A_a-module. The first term of the triangle is therefore weakly zero. The middle term is weakly zero by the preceding paragraph, so the relative associative cotangent tower $\{L^{\mathrm{ass}}_{B_a/A_a}\}_a$ is weakly zero. We use associative cotangent transitivity in the form of [15, Corollary 7.3.3.6 and Remark 7.4.1.12].

After restricting bimodule structures to K_a^{sp}, the ideal sequence [15, Proposition 7.4.1.15] gives

$$U_{\mathrm{sp}}(K_a^{\mathrm{an}}) \otimes_{K_a^{\mathrm{sp}}} J_a \simeq \Sigma^{-1} L^{\mathrm{ass}}_{U_{\mathrm{sp}}(K_a^{\mathrm{an}})/K_a^{\mathrm{sp}}}$$

and a fiber sequence

$$J_a \otimes_{K_a^{\mathrm{sp}}} J_a \longrightarrow J_a \longrightarrow U_{\mathrm{sp}}(K_a^{\mathrm{an}}) \otimes_{K_a^{\mathrm{sp}}} J_a.$$

Consequently,

$$\{J_a \otimes_{K_a^{\mathrm{sp}}} J_a\}_a \longrightarrow \{J_a\}_a$$

is a weak equivalence of towers. Repeated tensoring and [13, Lemma 2.29] give weak equivalences

$$\{J_a^{\otimes_{K_a^{\mathrm{sp}}} 2^r}\}_a \longrightarrow \{J_a\}_a \qquad (r \geq 0).$$

The sources are uniformly 2^{r+1}-connective. Given a fixed Postnikov degree n, choose r with $2^{r+1} > n$. Then $\{\tau_{\leq n} J_a\}_a \simeq 0$, which proves the weak equivalence of ring towers.

Now let S be a fixed connective central E_1-HR-algebra. Applying [13, Lemma 2.29] over the constant pro-ring HR gives

$$\{S \otimes_{HR} K_a^{\mathrm{sp}}\}_a \xrightarrow{\sim_{\mathrm{w}}} \{S \otimes_{HR} U_{\mathrm{sp}}(K_a^{\mathrm{an}})\}_a.$$

Centrality supplies the displayed tensor products with their E_1-ring structures; the connectivity bounds are unchanged.

Localizing invariants. Land–Tamme's connectivity criterion [13, Corollary 2.31 and Example 2.6] now gives the assertion for every ℓ-connective localizing invariant, both before and after the stated central scalar extension. □

8.2 Animated reflection

Relative animated reflection provides the bridge from a spectral algebra over a discrete base to an animated algebra while preserving the property of being locally almost of finite presentation.

Let A be discrete. Spectral Algebraic Geometry supplies an adjunction

$$L_A : \mathrm{CAlg}^{\mathrm{cn}}_{HA} \rightleftarrows \mathrm{CAlg}^{\Delta}_{A} : \Theta_A, \tag{8.3}$$

where Θ_A preserves limits and colimits and is conservative [17, Propositions 25.1.2.2 and 25.1.2.4]. It is compatible with Postnikov truncations [17, Remark 25.1.3.4].

Lemma 8.2 (Relative animated reflection)**.** *Let $HA \to B$ be a connective E_∞-HA-algebra almost of finite presentation. The unit*

$$\eta_B : B \longrightarrow \Theta_A L_A(B)$$

induces an isomorphism on π_0 and is almost of finite presentation. Consequently the induced morphism of affine spectral schemes is finite. The construction commutes with localization at elements.

Proof. The equivalence on 0-truncated objects [17, Remark 25.1.3.5], tested against discrete animated algebras, gives
$$\pi_0 L_A(B) \cong \pi_0 B.$$
To test whether $L_A(B)$ is almost of finite presentation, apply the adjunction (8.3) to truncated filtered diagrams. The functor Θ_A preserves their colimits and their Postnikov truncations, so the mapping-space criterion is inherited from B. The animated–spectral finiteness comparison of [12, Section 2.4] then shows that $\Theta_A L_A(B)$ is almost of finite presentation over HA. Cancellation for the property of being almost of finite presentation [17, Proposition 4.1.3.1] makes η_B almost of finite presentation. A morphism of affine spectral schemes almost of finite presentation which induces an isomorphism on π_0 is finite in this situation by [17, Remark 5.2.0.2].

For $f \in \pi_0 B$ and $g \in A$, the adjunction and the universal property of element localization give
$$L_A(B[f^{-1}]) \simeq L_A(B)[f^{-1}]$$
and
$$L_{A[g^{-1}]}(B \otimes_{HA} HA[g^{-1}]) \simeq L_A(B) \otimes_A A[g^{-1}].$$
These compatibilities prove the localization assertion. □

Corollary 8.3 (Geometric animated reflection)**.** *Let X be classical and let $Y \to X$ be a quasi-compact, separated morphism of connective spectral schemes which is locally almost of finite presentation. On every classical affine open* $\operatorname{Spec} A \subset X$*, choose a finite affine cover of its inverse image in Y, together with finite principal refinements of the overlaps. The affine reflections glue to a derived scheme Y^{an}, and the units glue to a finite morphism locally almost of finite presentation*
$$(Y^{\mathrm{an}})^{\mathrm{sp}} \longrightarrow Y.$$
The construction commutes with restriction to classical affine opens of X and induces an isomorphism on classical truncations.

Proof. Quasi-compactness supplies the finite affine covers, and separatedness makes their pairwise intersections quasi-compact. Refine these intersections by finitely many principal opens. The first localization identity in Lemma 8.2 identifies the reflections of two affine charts after restriction to every such principal open in Y. When the classical affine open in X is itself replaced by a principal open, the second identity identifies this restriction with reflection relative to the localized base. Both identifications arise from the adjunction (8.3) and the universal property of localization; they are therefore natural in chains of localizations. In particular, on triple intersections their two composites agree, and the unit maps satisfy the same cocycle condition. Zariski gluing gives Y^{an} and the displayed morphism, compatibly with restriction to classical affine opens of X. Finiteness and the property of being locally almost of finite presentation are affine-local on the target, and the π_0-identifications in the lemma give the assertion on classical truncations. □

8.3 Proper modifications over a classical base

The preceding two comparisons place the formal defect of a proper spectral modification over a classical base within the range of derived pro-cdh descent.

Proposition 8.4 (Proper pro-excision over a classical base)**.** *Let X be a classical qcqs scheme, regarded as a spectral scheme, and let $p : Y \to X$ be a proper morphism of connective spectral schemes which is locally almost of finite presentation. Suppose that p is* *an equivalence over a quasi-compact open $U \subset X$, and put $Z = |X| \setminus U$. Then the formal completion square*

$$\begin{tikzcd} E(X) \arrow[r] \arrow[d] & E(X^{\wedge}_{Z}) \arrow[d] \\ E(Y) \arrow[r] & E(Y^{\wedge}_{p^{-1}Z}) \end{tikzcd}$$

is weakly cartesian for $E = K$ and for $E = \mathrm{TC}$.

Proof. Animated reflection. Assume first that $X = \operatorname{Spec} A$. Since p is proper, Y is qcqs and separated. Corollary 8.3 gives a derived scheme Y^{an} and a morphism

$$q : (Y^{\mathrm{an}})^{\mathrm{sp}} \longrightarrow Y.$$

The affine construction shows that q is finite and locally almost of finite presentation. Over U the original morphism $Y \to X$ is an equivalence, and reflection fixes the relative initial algebra; hence q is also an equivalence over U.

The same charts show that the morphism $p^{\mathrm{an}} : Y^{\mathrm{an}} \to X$ is locally almost of finite presentation. Its classical truncation agrees with that of p. It is therefore proper by the truncation criterion [17, Remark 5.1.2.2], and it is an equivalence over U.

Relative formal terms and two-out-of-three. For a spectral $T \to X$, put

$$Z_T = |T| \setminus |T \times_X U|,$$

and let

$$\mathcal{R}^{\mathrm{sp}}_E(T) = \operatorname{fib}\big(E(T) \longrightarrow E(T^{\wedge,\mathrm{sp}}_{Z_T})\big),$$

where the superscript records that the completion is represented by the free spectral Koszul tower. If $V \to X$ is a derived scheme, define similarly

$$\mathcal{R}^{\mathrm{an}}_E(V) = \operatorname{fib}\big(E(V) \longrightarrow E(V^{\wedge,\mathrm{an}}_{Z_V})\big)$$

using the animated Koszul tower of [12, Section 2.5].

By [12, Section 2.2], modules, perfect complexes, and hence the value of a localizing invariant on a derived scheme agree with those on its underlying spectral scheme. Apply the derived pro-cdh theorem [12, Theorem 4.6] with base ring $k = \mathbb{Z}$ to p^{an}; its conclusion is the weak equivalence in the bottom row of the diagram below. Proposition 8.1 gives the left vertical comparison. On an affine chart of Y^{an}, the right vertical comparison is its central scalar extension along the underlying connective commutative HR-algebra. These affine comparisons globalize by the fixed finite binary Mayer–Vietoris presentation of Lemma 7.2. The construction of the ring-tower comparison is natural in the algebra and in localization, so the resulting diagram commutes:

$$\begin{tikzcd} \mathcal{R}^{\mathrm{sp}}_E(X) \arrow[r] \arrow[d, "\sim_{\mathrm{w}}"'] & \mathcal{R}^{\mathrm{sp}}_E((Y^{\mathrm{an}})^{\mathrm{sp}}) \arrow[d, "\sim_{\mathrm{w}}"] \\ \mathcal{R}^{\mathrm{an}}_E(X) \arrow[r, "\sim_{\mathrm{w}}"] & \mathcal{R}^{\mathrm{an}}_E(Y^{\mathrm{an}}). \end{tikzcd} \tag{8.4}$$

Thus the upper horizontal map is a weak equivalence.

Theorem 7.8, applied to the finite modification q, gives

$$\mathcal{R}^{\mathrm{sp}}_E(Y) \xrightarrow{\sim_{\mathrm{w}}} \mathcal{R}^{\mathrm{sp}}_E((Y^{\mathrm{an}})^{\mathrm{sp}}). \tag{8.5}$$

Since $p^{\mathrm{an}} = p \circ q$ after passage to underlying spectral schemes, naturality identifies the upper horizontal map in (8.4) with the composite of $\mathcal{R}_E^{\mathrm{sp}}(X) \to \mathcal{R}_E^{\mathrm{sp}}(Y)$ and (8.5). Two-out-of-three therefore makes the first map a weak equivalence. Its fiber is the total fiber of the formal square in the statement, proving the affine-base case.

Globalization. For a general qcqs classical base, choose for X the finite binary affine presentation furnished by Lemma 7.2, together with finite affine covers of its proper inverse images and finite principal refinements of their intersections. The affine argument applies on every member of this fixed finite system. Zariski descent for K and TC expresses the global comparison by finitely many pullbacks. The number of these operations is independent of the formal parameter, so every fixed Postnikov truncation of the local weak pro-equivalences remains an equivalence after globalization. □

Thus $\Phi_E(p)$ is weakly zero over a classical base. The next two sections show that the formal defect is unchanged by passage from a spectral base to its classical truncation.

9 Cyclotomic coefficients and proper formal functions

To prove 1-connective invariance of the formal defect, the next section reduces to a square-zero base change. Relative TC for such a layer has a weight filtration whose graded pieces are homotopy orbits of THH with central coefficients. This section proves that each coefficient defect vanishes along the completion tower. Two inputs are needed: finite-group orbits preserve uniformly bounded weakly pro-zero towers, and central coefficient THH satisfies proper formal functions.

9.1 Finite-group orbits

Lemma 9.1 (Finite-group orbits and weak pro-zero towers)**.** *Let G be finite, and let $\{V_a\}$ be a pro-object of* $\mathrm{Fun}(BG, \mathrm{Sp}_{\geq c})$ *for a fixed integer c. If its underlying pro-spectrum is weakly zero, then $\{(V_a)_{hG}\}$ is weakly zero. Consequently, homotopy orbits preserve weak equivalences between uniformly bounded-below pro-G-spectra.*

Proof. Fix an output degree k and a stage of the tower. Choose $N > k - c$. Through degree k, the bar realization computing $(V_a)_{hG}$ agrees with its N-skeleton: every cell above the N-skeleton is at least $(c+N+1)$-connective. The skeletal filtration is finite, and its graded pieces are finite coproducts of suspensions $\Sigma^s V_a$, for $0 \leq s \leq N$.

Because the underlying tower is weakly zero, we may choose successively $N+1$ later transitions that vanish after all truncations required by these graded pieces. Each induced map on truncated N-skeleta lowers the skeletal filtration by one. Their composite is therefore zero. This proves that $\{\tau_{\leq k}(V_a)_{hG}\}$ is pro-zero. Applying the same argument to the cofiber of a weak equivalence proves the final assertion. □

The bounded-below hypothesis makes the finite-skeleton argument possible. This lemma concerns homotopy orbits; homotopy fixed points and Tate constructions do not have the same finite approximation in a fixed output degree.

9.2 Square-zero weights

We next identify the weight pieces to which the lemma will be applied. For an E_1-ring S and an S-bimodule N, write

$$\mathrm{THH}(S; N) := \left|[q] \longmapsto N \otimes S^{\otimes q}\right| \simeq N \otimes_{S \otimes S^{\mathrm{op}}} S$$

for coefficient topological Hochschild homology; the faces in the displayed cyclic-bar construction use the two S-actions on N and multiplication in S. In the square-zero situation below, the ideal M is naturally an S-bimodule.

Lemma 9.2 (Integral nonsplit square-zero filtration)**.** *Let $R \to S$ be a square-zero extension of connective E_1-rings, and write $M = \mathrm{fib}(R \to S)$. The relative spectrum*

$$\mathrm{TC}(R/S) = \mathrm{fib}(\mathrm{TC}(R) \longrightarrow \mathrm{TC}(S))$$

has a natural complete decreasing filtration by positive weights, with

$$\mathrm{gr}^m \, \mathrm{TC}(R/S) \simeq \mathrm{THH}(S; (\Sigma M)^{\otimes_S m})_{hC_m}, \qquad m \geq 1. \tag{9.1}$$

If M is n-connective, the weight-m graded piece is at least $m(n+1)$-connective, while its tail $F^m\mathrm{TC}(R/S)$ is at least $(m(n+1)-1)$-connective. The filtration is natural for maps of square-zero extensions. In particular, it is natural under scalar extension once the scalar-extension comparison has been identified independently as a map of square-zero extensions with the base-changed ideal.

Proof. Put

$$V = \mathrm{THH}(R/S) = \mathrm{fib}(\mathrm{THH}(R) \longrightarrow \mathrm{THH}(S)).$$

Give the nonsplit extension its two-step deformation filtration, reindexed by positive weight. In Raskin's convention its associated graded has $\mathrm{gr}_0 = S$ and $\mathrm{gr}_{-1} = M$. The filtered cyclotomic construction of [22, Example 5.18.5 and Lemma 5.19.1] gives

$$\mathrm{gr}\, V \simeq \mathrm{THH}(S \oplus M\{1\}/S), \qquad \varphi_p(F^m V) \subseteq F^{pm}(V^{tC_p}),$$

and its tails become increasingly connective. Here $\{1\}$ records filtration weight; the suspension in $(\Sigma M)^{\otimes m}$ instead comes from the split square-zero trace calculation.

For the cyclotomic spectrum V, the integral equalizer is initially formed with profinitely completed periodic cyclic homology:

$$\mathrm{TC}(V) = \mathrm{Eq}\left(\mathrm{TC}^-(V) \underset{\varphi}{\overset{\mathrm{can}}{\rightrightarrows}} \widehat{\mathrm{TP}}(V)\right), \tag{9.2}$$

where $\widehat{\mathrm{TP}}(V)$ denotes the profinite completion of $\mathrm{TP}(V) = V^{t\mathbb{T}}$; this is [22, Equation (5.16.1)], following [20, Remark II.4.3].

In the present square-zero situation, Lemma 5.19.1 of [22] verifies all the hypotheses of Proposition 5.18.6 there. The conclusion is that V admits a meromorphic Frobenius; in particular, the canonical comparison

$$\mathrm{TP}(V) \xrightarrow{\sim} \widehat{\mathrm{TP}}(V)$$

is an equivalence. Step 1 of the proof of that proposition supplies complete filtrations on $V^{h\mathbb{T}} = \mathrm{TC}^-(V)$, $V_{h\mathbb{T}}$, and $\mathrm{TP}(V)$, and the preceding comparison is an equivalence of filtered spectra after transporting the filtration to the completed term. Thus (9.2) may be rewritten as

$$\mathrm{TC}(V) = \mathrm{Eq}\left(\mathrm{TC}^-(V) \underset{\varphi}{\overset{\mathrm{can}}{\rightrightarrows}} \mathrm{TP}(V)\right).$$

This two-step argument is what ensures that the displayed equalizer computes integral, rather than only p-typical, TC.

The functor TC on cyclotomic spectra is representable and hence preserves limits; since its source and target are stable, it is exact. It follows that

$$\mathrm{TC}(V) \simeq \mathrm{fib}(\mathrm{TC}(R) \longrightarrow \mathrm{TC}(S)) = \mathrm{TC}(R/S).$$

Taking the preceding equalizer levelwise defines $F^m\mathrm{TC}(V)$. Complete filtered spectra are closed under finite limits, so this filtration is complete. Moreover, associated graded is exact and therefore commutes with this finite equalizer.

On the weight-m quotient, can preserves weight. For every prime p and every $m \geq 1$, the filtered Frobenius satisfies

$$\varphi_p(F^m\mathrm{TC}^-(V)) \subseteq F^{pm}\mathrm{TP}(V) \subseteq F^{m+1}\mathrm{TP}(V)$$

by [22, Lemma 4.8.1]. Hence every prime component, and therefore the amalgamated integral Frobenius, induces the zero map on gr^m:

$$\mathrm{gr}^m(\varphi) = 0.$$

It follows that

$$\mathrm{gr}^m\,\mathrm{TC}(V) \simeq \mathrm{fib}\left((\mathrm{gr}^m V)^{h\mathbb{T}} \xrightarrow{\mathrm{can}} (\mathrm{gr}^m V)^{t\mathbb{T}}\right).$$

Raskin's split trace formula [22, Proposition 4.5.1] identifies $\mathrm{gr}^m V$ with

$$\mathrm{Ind}_{C_m}^{\mathbb{T}}\,\mathrm{THH}(S; (\Sigma M)^{\otimes_S m}).$$

Here $\mathrm{Ind}_{C_m}^{\mathbb{T}}$ denotes Raskin's right-adjoint induction functor (equivalently, coinduction), not left-adjoint induction. The norm-cofiber calculation [22, Lemma 4.9.1 and Theorem 4.10.1] now gives (9.1). The deformation filtration, filtered THH, the equalizer, and the cyclic trace construction are functorial for morphisms of square-zero extensions, proving naturality. For an external scalar extension, exactness identifies the new ideal with the base-changed ideal, which is the scalar-extension case stated in the lemma.

If M is n-connective, then $(\Sigma M)^{\otimes_S m}$ is $m(n+1)$-connective. Coefficient THH and finite-group homotopy orbits are left t-exact, so the graded piece $\mathrm{gr}^m\,\mathrm{TC}(R/S)$ satisfies this bound. This is a bound on the graded piece, not yet on the tail. For $N > m$, the quotient F^m/F^N is a finite extension of weights $m, \ldots, N-1$. Completeness writes F^m as the inverse limit of these finite quotients, and the Milnor exact sequence loses at most one degree. Thus the whole weight-m tail is $(m(n+1)-1)$-connective, uniformly in any diagram in which the lower bound for M is uniform. □

The construction is integral: it derives the filtration directly from Raskin's filtered cyclotomic formalism, while Lee–Levy's parallel nonsplit argument is written under their global p-complete convention [14, Conventions].

9.3 Central coefficients and proper formal functions

Lemma 9.3 (Almost finite presentation of derived loop maps)**.** *Let $f : Y \to X$ be a morphism of connective spectral schemes which is locally almost of finite presentation, and put $L(-) = \mathrm{Map}(S^1, -)$. The derived loop spaces LY and LX are connective spectral schemes, and the induced morphism $Lf : LY \to LX$ is locally almost of finite presentation. More precisely, if $\gamma : T \to LY$ has adjoint $\widetilde{\gamma} : S^1 \times T \to Y$, then*

$$\mathbb{L}_{LY/LX,\gamma} \simeq \pi_!\widetilde{\gamma}^*\mathbb{L}_{Y/X}, \qquad \pi : S^1 \times T \to T.$$

If f is proper, then Lf is proper.

Proof. Connective spectral schemes admit finite fiber products, and finite fiber products remain connective. Consequently the finite derived self-intersections

$$LY = Y \times_{Y\times Y} Y, \qquad LX = X \times_{X\times X} X$$

are connective spectral schemes. Let $\gamma : T \to LY$ and $\widetilde{\gamma} : S^1 \times T \to Y$ be adjoint maps. For every T-module N, relative derivations and the adjunction $\pi_! \dashv \pi^*$ identify

$$\mathrm{map}_{\mathrm{QCoh}(S^1\times T)}(\widetilde{\gamma}^*\mathbb{L}_{Y/X}, \pi^* N) \simeq \mathrm{map}_{\mathrm{QCoh}(T)}(\pi_!\widetilde{\gamma}^*\mathbb{L}_{Y/X}, N).$$

Thus the displayed object corepresents the relative cotangent functor. For S^1, a finite cellular model computes $\pi_!$ by a finite cofiber of almost-perfect T-modules (equivalently, by the cofiber of monodromy minus the identity). It therefore preserves almost-perfect objects. Proposition 17.1.5.1(3) of [17] shows that $\mathbb{L}_{Y/X}$ is almost perfect, and hence so is $\mathbb{L}_{LY/LX}$.

It remains to justify the converse finiteness implication. Functors of points represented by spectral schemes are infinitesimally cohesive, and this property is preserved by the finite limits defining LY and LX. On discrete rings, a scheme has discrete mapping spaces, so evaluation at the base point gives $LY|_{\mathrm{CAlg}^\heartsuit} \simeq Y|_{\mathrm{CAlg}^\heartsuit}$ and similarly for X. Thus Lf satisfies condition $(*)$ of Proposition 17.4.2.1 in [17], because the classical truncation $t_0 f$ is locally of finite presentation. The converse direction of [17, Corollary 17.4.2.2], followed by [17, Proposition 17.4.3.1], now shows that Lf is locally almost of finite presentation. Finally $t_0 LY \simeq t_0 Y$, $t_0 LX \simeq t_0 X$, and properness is detected on classical truncations by [17, Remark 5.1.2.2]. $\square$

For a connective spectral scheme X, let

$$LX = \mathrm{Map}(S^1, X) = X \times_{X\times X} X, \qquad e_X : LX \longrightarrow X$$

be its derived loop space and evaluation map. For a central coefficient $P \in \mathrm{QCoh}(X)$, define

$$\mathrm{THH}(X; P) := \Gamma(LX, e_X^* P). \tag{9.3}$$

When X is connective and qcqs, this loop-space definition agrees with the trace in presentable categories:

$$\mathrm{THH}(X; P) \simeq \mathrm{Tr}_{\mathrm{Pr}^L}(- \otimes P). \tag{9.4}$$

When P is not perfect, the right side is not a trace of an endofunctor of $\mathrm{Perf}(X)$. Indeed, for connective qcqs X, compact generation [17, Proposition 9.6.1.1] and the tensor-kernel formalism [17, Proposition 9.4.3.1 and Corollaries 9.4.2.4 and 9.4.4.6] identify $- \otimes P$ with the kernel $\Delta_* P$. Its trace is $\Gamma(X, \Delta^*\Delta_* P)$, which gives (9.4); see also [3]. On an affine this is the cyclic-bar identity $P \otimes_{A\otimes A} A$.

The definition (9.3), and in the qcqs case the trace identification (9.4), concern underlying spectra and do not, for an arbitrary P, supply a circle action. For a cyclic word, use the marked-point model. Let $e_{i/m} : LX \to X$ be evaluation at the point $i/m \in S^1$ and put

$$\mathcal{W}_m(M) = \bigotimes_{i\in\mathbb{Z}/m} e_{i/m}^* M \in \mathrm{QCoh}(LX). \tag{9.5}$$

Rotation by $1/m$, together with cyclic permutation of the factors, gives $\mathcal{W}_m(M)$ a canonical coherent C_m-linearization. After forgetting the action, the evaluation maps are homotopic and

$$\mathcal{W}_m(M) \simeq e_0^*(M^{\otimes m}).$$

Thus $\Gamma(LX, \mathcal{W}_m(M))$ is the usual coefficient trace as an underlying spectrum, equipped with its cyclic-word action. Loop maps, pullback, proper pushforward, base change, and the projection formula respect the marked points and this action. On an affine, the cyclic-bar model identifies this linearization with the cyclic-word C_m-action in Raskin's split trace formula.

Write

$$\mathrm{THH}_m(X; M) := \Gamma(LX, \mathcal{W}_m(M)) \in \mathrm{Fun}(BC_m, \mathrm{Sp}).$$

Lemma 9.4 (Equivariance of cyclic-word coefficient comparisons)**.** *Let $p : Y \to X$ be a proper morphism of connective qcqs spectral schemes, locally almost of finite presentation. Let $q = Lp$, and let $M \in \mathrm{QCoh}(X)$. Write $\mathrm{QCoh}(LX)^{hC_m}$ for the category of quasi-coherent sheaves equivariant for the rotation action. The morphism q is equivariant for loop rotation, and*

$$D_p := \mathrm{fib}(\mathcal{O}_{LX} \longrightarrow q_*\mathcal{O}_{LY})$$

is canonically an object of $\mathrm{QCoh}(LX)^{hC_m}$. The unit, base-change, and projection-formula maps give a natural equivalence in $\mathrm{Fun}(BC_m, \mathrm{Sp})$

$$\mathrm{fib}(\mathrm{THH}_m(X; M) \longrightarrow \mathrm{THH}_m(Y; p^*M)) \simeq \Gamma(LX, \mathcal{W}_m(M) \otimes D_p). \tag{9.6}$$

*It is functorial for base change in X. In particular, for a tower $\{X_a \to X\}$ and its pullback to Y, the central-coefficient formal square, with coefficients $M^{\otimes m}$, is a square in $\mathrm{Fun}(BC_m, \mathrm{Sp})$ whose underlying square is the usual coefficient-*THH *square.*

Proof. Use the cyclic bar model with the m marked points of the circle. Rotation of a loop by $1/m$ and cyclic permutation of the marked tensor factors give $\mathcal{W}_m(M)$ its C_m-linearization. The map Lp commutes with rotation, and the unit $\mathcal{O}_{LX} \to q_*\mathcal{O}_{LY}$ is a map of equivariant quasi-coherent sheaves. Thus D_p and $\mathcal{W}_m(M) \otimes D_p$ are equivariant. Pullback, proper pushforward, the base-change transformation, and the projection formula are natural transformations of the corresponding functors on the cyclic bar diagrams; hence they are C_m-equivariant. The equivariant global-sections functor

$$\Gamma(LX, -) : \mathrm{QCoh}(LX)^{hC_m} \longrightarrow \mathrm{Fun}(BC_m, \mathrm{Sp})$$

sends this comparison to (9.6); no homotopy fixed points are being taken. The same cyclic diagram is natural in every map $X_a \to X$, which proves the tower assertion. After forgetting the action, all evaluation maps are homotopic and the construction reduces to the standard coefficient trace $\mathrm{THH}(X; M^{\otimes m})$. □

Lemma 9.5 (Derived loops and formal completion)**.** *Let X be a connective affine spectral scheme, let $Z \subset |X|$ be closed with quasi-compact complement, and let $\{X_a\}$ be a cofinal affine presentation of $X^\wedge_Z$. Then there is a canonical equivalence of formal prestacks*

$$L(X^\wedge_Z) \simeq (LX)^\wedge_{e_X^{-1}Z}. \tag{9.7}$$

The tower $\{L(X_a)\}_a$ presents the right side. If $\{K_b\}_b$ is a free spectral Koszul tower on LX cutting out $e_X^{-1}Z$, then $\{L(X_a)\}_a$ and $\{K_b\}_b$ represent isomorphic pro-affine objects. After replacing the indexing categories by a common cofiltered category, this pro-isomorphism may be represented by a level map. Its level maps are not asserted to be equivalences.

Proof. The finite simplicial set S^1 is compact, so pointwise filtered colimits of spaces commute with $\mathrm{Map}(S^1, -)$. It follows that

$$\operatorname*{colim}_a L(X_a) \simeq L(\operatorname*{colim}_a X_a) = L(X^\wedge_Z).$$

It remains to identify this formal prestack. Let T be a connective affine test scheme. By the functor-of-points description of formal completion, a map $T \to X^\wedge_Z$ is a map $T \to X$ whose restriction to T_{red} factors through Z. The space $X(T_{\mathrm{red}})$ is discrete. Hence every loop in it is constant, and a map $S^1 \times T \to X$ factors through $X^\wedge_Z$ precisely when its value at one, equivalently every, point of S^1 lies formally along Z. This is exactly the condition defining $(LX)^\wedge_{e_X^{-1}Z}(T)$, which proves (9.7).

Both displayed towers are Ind-affine presentations of this prestack. Our convention is that an Ind-affine presentation is a filtered colimit of affine spectral schemes in prestacks; contravariance of Spec associates to it a pro-object of connective E_∞-rings. Full faithfulness of the Ind-affine Yoneda embedding identifies the associated pro-affine objects. The standard reindexing theorem for pro-objects then supplies a common cofiltered indexing category on which the resulting pro-isomorphism is represented by a level map. Reindexing does not make the individual level maps equivalences. ☐

Lemma 9.6 (Proper formal functions for central THH coefficients). *Let $p : Y \to X$ be proper and locally almost of finite presentation between connective qcqs spectral schemes. Suppose that p is an equivalence over a quasi-compact open $U \subset X$, and put $Z = |X|\setminus U$. Let $P \in \mathrm{QCoh}(X)$ be bounded below. For the fixed cofinal Koszul tower $\{X_a\}$, put $Y_a = Y \times_X X_a$ and $P_a = P|_{X_a}$. Then the central-coefficient formal square is*

$$\begin{array}{ccc} \mathrm{THH}(X;P) & \longrightarrow & \{\mathrm{THH}(X_a;P_a)\}_a \\ \Big\downarrow & & \Big\downarrow \\ \mathrm{THH}(Y;p^*P) & \longrightarrow & \{\mathrm{THH}(Y_a;p_a^*P_a)\}_a. \end{array}$$

This square is weakly cartesian. If $M \in \mathrm{QCoh}(X)$ is bounded below and $P = (\Sigma M)^{\otimes_{\mathcal{O}_X} m}$, its cyclic-word refinement is a square in $\mathrm{Fun}(BC_m, \mathrm{Sp})$ by Lemma 9.4. Its underlying stagewise total fibers have a lower bound independent of the formal index, and the square remains weakly cartesian after homotopy C_m-orbits.

Proof. Loop geometry. We first work over an affine X; a finite affine Mayer–Vietoris argument will then give the general case. Put $q = Lp : LY \to LX$. The loop construction is a finite derived self-intersection, so q is qcqs. Lemma 9.3 shows that it is proper and locally almost of finite presentation.

The supported coefficient defect. Let

$$D = \mathrm{fib}(\mathcal{O}_{LX} \longrightarrow q_*\mathcal{O}_{LY}).$$

The proper direct-image theorem [17, Theorem 5.6.0.2] makes D almost perfect. It vanishes over LU, so it is supported on $e_X^{-1}Z$. Arbitrary derived base change and the projection formula [17, Corollary 3.4.2.2(3) and Remark 3.4.2.6] identify the fiber of the coefficient comparison with

$$\Gamma(LX, e_X^*P \otimes D). \tag{9.8}$$

Completion on the loop space. Choose a cofinal tower $\{X_a\}$ for the completion of X along Z, and put $Y_a = Y \times_X X_a$. Preservation of limits by the loop functor gives

$$L(Y_a) = LY \times_{LX} L(X_a).$$

Lemma 9.5 identifies $\{L(X_a)\}$ with the formal completion of LX along $e_X^{-1}Z$. Choose a free spectral Koszul tower $\{K_b\}$ on the affine scheme LX for this center; here LX is affine because X is affine. The same lemma gives an isomorphism between the pro-affine objects represented by $\{K_b\}$ and $\{L(X_a)\}$. After passage to a common cofiltered index, we use a level-map representative of this pro-isomorphism; its components need not be equivalences.

Write $j_a : L(X_a) \to LX$ and $q_a = L(p_a)$. Derived base change yields

$$D_a := \mathrm{fib}(\mathcal{O}_{L(X_a)} \longrightarrow (q_a)_*\mathcal{O}_{L(Y_a)}) \simeq j_a^*D.$$

Choose fixed suspensions which make both D and e_X^*P connective. Since D is supported on $e_X^{-1}Z$, Lemma 7.7, applied first to the free tower $\{K_b\}$, identifies it weakly with its completion tower. The tensor-stability theorem [13, Lemma 2.29] preserves this comparison after tensoring with the suspended e_X^*P. Desuspending and transporting the comparison across the preceding pro-isomorphism gives

$$\Gamma(LX, e_X^*P \otimes D) \xrightarrow{\sim_{\mathrm{w}}} \{\Gamma(L(X_a), j_a^*(e_X^*P \otimes D))\}_a.$$

Derived base change and the projection formula identify the terms on the right with the stagewise analogues of (9.8). This is precisely the weak cartesianness of the coefficient formal square. In Section 10, these coefficient defects occur as the graded pieces of the TC formal defect.

Equivariance and globalization. Let $M \in \mathrm{QCoh}(X)_{\geq c_M}$ and use the marked coefficient $\mathcal{W}_m(\Sigma M)$. Lemma 9.4 places the entire four-corner formal square, including its comparison map, in $\mathrm{Fun}(BC_m, \mathrm{Sp})$. The preceding argument proves that its underlying total-fiber tower is weakly zero.

We record the uniform bound used here. Since D is almost perfect on the affine spectral scheme LX, choose c_D with $D \in \mathrm{QCoh}(LX)_{\geq c_D}$. Pullback along every connective formal stage preserves these bounds, and

$$\mathcal{W}_m(\Sigma M) \otimes D \in \mathrm{QCoh}(LX)_{\geq m(c_M+1)+c_D}.$$

The same estimate holds after pullback to every $L(X_a)$. If d_0 is the fixed number of successive fiber operations used to form the affine formal total fiber, then every stage of its underlying tower lies in

$$\mathrm{Sp}_{\geq m(c_M+1)+c_D-d_0};$$

in particular, for fixed m this lower bound is independent of the formal index a. Lemma 9.1 therefore applies to its homotopy C_m-orbits. Notice that this is a bound for the weight-m graded coefficient defect; the increasing bound for the full tail F^m is the separate estimate in Lemma 9.2.

Finally choose one fixed finite affine Mayer–Vietoris presentation. The loop-space coefficient traces satisfy the resulting finite Zariski descent, and finite limits preserve the equivariant comparisons. After replacing c_D by the minimum of the finitely many chartwise bounds, let d_{MV} be the number of additional fiber operations in this fixed presentation. The global stagewise total fibers are bounded below by

$$m(c_M+1) + c_D - d_0 - d_{\mathrm{MV}},$$

still independently of a. This globalizes both the underlying and the C_m-orbit statements. □

10 1-connective invariance of the formal defect

We now prove that the formal defect is invariant under a 1-connective change of the base. There are two independent uniformity arguments. A complete weight filtration removes a square-zero layer, while the Lee–Levy factorization replaces the remainder by maps of exponentially increasing connectivity. Both arguments are performed on the finite four-vertex formal defect itself; no infinite Čech totalization is interchanged with a Postnikov truncation.

For connective qcqs spectral schemes, call a morphism $p : Y \to X$ a U-modification if it is proper and locally almost of finite presentation and is an equivalence over the quasi-compact open $U \subset X$.

Lemma 10.1 (Lee–Levy factorization with external base change)**.** *Let $f : R \to S$ be an n-connective map of connective commutative A-algebras, where $n \geq 1$, and form*

$$C(f) = S \times_{S \otimes_R S} S$$

in commutative ring spectra. There are natural maps

$$R \xrightarrow{h_f} C(f) \xrightarrow{g_f} S$$

such that h_f is $2n$-connective and g_f is a central, possibly nonsplit, n-connective square-zero extension. Moreover, for every connective commutative A-algebra A', the canonical map

$$C(f) \otimes_A A' \longrightarrow C(R \otimes_A A' \longrightarrow S \otimes_A A')$$

is an equivalence and identifies the square-zero ideals.

Proof. After forgetting to E_1-rings, the displayed pullback is the algebra of [14, Construction 7.10]: limits of commutative algebras are created on underlying E_1-algebras. The connectivity assertions, the square-zero structure, and the identity

$$\mathrm{fib}(h_f) \simeq \mathrm{fib}(f) \otimes_R \mathrm{fib}(f)$$

are [14, Lemma 7.11]. Since $C(f)$ and both structure maps were formed in commutative algebras, the square-zero ideal is a central S-module.

For base change, the underlying-module functor preserves limits, while $- \otimes_A A'$ is exact between stable module categories and hence preserves finite pullbacks. Associativity gives

$$(S \otimes_R S) \otimes_A A' \simeq (S \otimes_A A') \otimes_{R \otimes_A A'} (S \otimes_A A').$$

Writing U_A and $U_{A'}$ for the underlying-module functors, every commutative A-algebra B has

$$U_{A'}(B \otimes_A A') \simeq U_A(B) \otimes_A A' \quad \text{in } \mathrm{Mod}_{A'}.$$

Consequently the canonical base-change map is an equivalence on underlying modules, hence an equivalence of commutative algebras. Naturality also identifies h_f, g_f, and the fiber of g_f. □

Lemma 10.2 (Filtered pro-dévissage)**.** *Let $V = \{V_a\}$ be a pro-spectrum with a natural decreasing filtration $F^1 V_a = V_a$. Suppose that, for every fixed $m \geq 1$, the pro-spectrum $\{\mathrm{gr}^m V_a\}_a$ is weakly zero. Suppose also that there is a function $b(m) \to +\infty$ for which*

$$F^m V_a \in \mathrm{Sp}_{\geq b(m)} \qquad \textit{for every } a.$$

Then V is weakly zero.

Proof. Fix k and choose M with $b(M) > k$. The spectrum $F^M V_a$ has zero k-truncation for every a. The quotient $V_a / F^M V_a$ has a finite filtration with graded pieces $\mathrm{gr}^1 V_a, \ldots, \mathrm{gr}^{M-1} V_a$. Weakly zero pro-spectra are closed under finite extensions, so $\{\tau_{\leq k} V_a\}_a$ is zero. Since k was arbitrary, V is weakly zero. □

Let $v : X' \to X$ be a base map, put $Y' = Y \times_X X'$, and pull the fixed formal tower strictly to X' and Y'. For an invariant E, write

$$\Delta_E(v; p) = \mathrm{fib}(\Phi_E(p) \longrightarrow \Phi_E(p')). \tag{10.1}$$

By stability, this is the total fiber of the formal four-vertex square whose entries are the vertical relative spectra $\mathrm{fib}(E(T) \to E(T \times_X X'))$ for $T = X, Y, X_a, Y_a$.

Lemma 10.3 (The Lee–Levy square-zero layer)**.** *Let $f : R \to S$ be an n-connective map of connective commutative rings, with $n \geq 1$, and put $C = C(f)$. Let $p_R : Y_R \to \operatorname{Spec} R$ be a U-modification, and let p_C, p_S be its base changes to $\operatorname{Spec} C, \operatorname{Spec} S$. Obtain all formal stages by external scalar extension from the fixed tower over R. Then*

$$\Phi_{\mathrm{TC}}(p_C) \longrightarrow \Phi_{\mathrm{TC}}(p_S)$$

is a weak equivalence.

Proof. Choose once and for all a finite affine Mayer–Vietoris presentation for Y_R, and pull it to Y_C, Y_S and every formal stage. Let d bound the number of successive fiber operations needed to form these presentations and the four-vertex total fiber. The same finite diagrams are used for every weight and every external scalar extension; hence d is independent of the formal index a and the weight m. When this lemma is applied to the iteration in Theorem 10.4, the same choice is also independent of the iteration index r. Since $R \to C \to S$ consists of π_0-isomorphisms and free commutative quotients commute with scalar extension, the pulled-back towers over C and S are the free Koszul towers on the images of the original generators and are cofinal presentations of the required formal centers.

Write $g : C \to S$ for the square-zero factor and $M = \operatorname{fib}(g) \in \mathrm{Mod}_{S \geq n}$. Every affine vertex in the fixed diagrams is obtained from R by a connective external scalar extension $R \to B$. Lemma 10.1 identifies

$$C \otimes_R B \longrightarrow S \otimes_R B$$

with the Lee–Levy square-zero factor of $B \to S \otimes_R B$. Put

$$B_S = S \otimes_R B, \qquad M_B = M \otimes_S B_S \simeq M \otimes_R B \in \mathrm{Mod}_{B_S \geq n}.$$

The ideal at this vertex is M_B. Thus every vertex is again a central square-zero extension with an n-connective ideal of the correct coefficient-ring type. All square-zero extensions used below therefore arise by the controlled external scalar extensions covered by Lemma 10.1.

Put

$$V = \operatorname{fib}(\Phi_{\mathrm{TC}}(p_C) \longrightarrow \Phi_{\mathrm{TC}}(p_S)).$$

Apply Lemma 9.2 at every vertex. Naturality gives the vertical relative TC square, and finite total fibers give V a complete filtration: complete filtered spectra form a stable subcategory closed under finite limits. Its weight-m graded piece is the total fiber of the coefficient formal square which, at the vertex B_S, has entry

$$\mathrm{THH}\big(B_S; (\Sigma M_B)^{\otimes_{B_S} m}\big)_{hC_m}. \tag{10.2}$$

Equivalently, regard M as an object of $\mathrm{QCoh}(\operatorname{Spec} S)_{\geq n}$; then (10.2) is the cyclic-word coefficient formal square for the base-changed modification over $\operatorname{Spec} S$.

Lemma 9.6, with its marked-point cyclic-word linearization, makes the underlying coefficient square weakly cartesian and uniformly bounded below. Lemma 9.1 therefore permits homotopy C_m-orbits. Since homotopy orbits are exact, they commute with the finite total fiber. Thus $\mathrm{gr}^m V$ is weakly zero for every fixed m.

The tail estimate in Lemma 9.2, followed by the fixed finite diagram, gives the levelwise bound

$$F^m V_a \in \mathrm{Sp}_{\geq m(n+1)-1-d} \qquad \text{for every } a. \tag{10.3}$$

The right side tends to $+\infty$ with m, uniformly in the formal index. Lemma 10.2 proves the claim. ☐

Theorem 10.4 (Invariance under 1-connective affine base change)**.** *Let $p : Y \to X$ be a U-modification of connective qcqs spectral schemes. Let $v : X' \to X$ be affine, where X' is a connective qcqs spectral scheme. Assume that there is an integer $n \geq 1$ such that, for every affine open $W = \operatorname{Spec} R \subset X$, one has $X' \times_X W \simeq \operatorname{Spec} S$ and the map of connective commutative ring spectra $R \to S$ is n-connective. Put $Y' = Y \times_X X'$, write $p' : Y' \to X'$, and obtain every formal stage by strict base change from the fixed tower on X. Then*

$$\Phi_E(p) \longrightarrow \Phi_E(p')$$

is a weak equivalence for $E = \mathrm{TC}$ and for nonconnective K-theory.

Proof. Since $n \geq 1$, the map v induces an equivalence on classical truncations. The affine hypothesis presents v on every affine open $W = \operatorname{Spec} R \subset X$ by the specified map $R \to S$. Lemma 10.1 is compatible with localization; hence its local factorizations glue and commute with base change to Y and to every formal stage. Since all intermediate maps are π_0-isomorphisms and free commutative cell attachments commute with scalar extension, strict pullback of $R//(f_1^a, \dots, f_s^a)$ is the corresponding free Koszul stage over every intermediate ring. Thus the pulled-back towers remain cofinal presentations of the same formal center.

Equivalently, this gluing may be expressed globally. Regard $v_*\mathcal{O}_{X'}$ as a commutative algebra in $\mathrm{QCoh}(X)$ and form

$$\mathcal{C} = v_*\mathcal{O}_{X'} \underset{v_*\mathcal{O}_{X'} \otimes_{\mathcal{O}_X} v_*\mathcal{O}_{X'}}{\times} v_*\mathcal{O}_{X'}.$$

Its restriction to every affine chart is $C(f)$, so its relative spectrum is the global first Lee–Levy intermediate. Repeating this construction globalizes every finite stage of the iteration below.

We first treat TC. On an affine write $f_0 : R \to S^{(0)} = S$ for the map defining v. Recursively factor

$$f_r : R \xrightarrow{f_{r+1}} S^{(r+1)} = C(f_r) \xrightarrow{g_r} S^{(r)}. \tag{10.4}$$

Then f_r is $2^r n$-connective and g_r is a central $2^r n$-connective square-zero extension. Let $X^{(r)} \to X$ be the resulting global thickening and let $p^{(r)}$ be the base change of p. Before applying the square-zero lemma, fix finite affine Mayer–Vietoris presentations for the four vertices of the formal defect and pull them through every external scalar extension $X^{(r)} \to X$. On each affine term, Lemma 10.3, applied to f_r and all its external scalar extensions in this fixed diagram, gives a weak equivalence. Zariski descent for TC expresses the global formal defects by the same finite limits. Since weak equivalences of pro-spectra are preserved by finite limits, the chartwise equivalences globalize to

$$\Phi_{\mathrm{TC}}(p^{(r+1)}) \xrightarrow{\sim_{\mathrm{w}}} \Phi_{\mathrm{TC}}(p^{(r)}). \tag{10.5}$$

Consequently the comparison fiber for $X' \to X$ is weakly equivalent, for every finite r, to the comparison fiber for $X^{(r)} \to X$.

Using this same fixed presentation, let d bound the number of successive fiber operations in the presentations together with the final four-vertex total fiber. The diagram is fixed before choosing a or r, so d is independent of the formal index a and the iteration index r. Connectivity is preserved by every connective scalar extension, so all affine vertical maps at the four vertices and every formal stage remain $2^r n$-connective. Integral TC is a 1-connective localizing invariant [13, Example 2.6]. Hence every affine vertical relative TC term is at least $(2^r n + 1)$-connective. Taking the fixed finite presentations and the four-vertex total fiber gives

$$\Delta_{\mathrm{TC}}(X^{(r)} \to X; p)_a \in \mathrm{Sp}_{\geq 2^r n + 1 - d} \qquad \text{for every formal stage } a. \tag{10.6}$$

Given k, choose a finite r with $2^r n + 1 - d > k$. Equations (10.5) and (10.6) show that the k-truncation of $\Delta_{\mathrm{TC}}(v;p)$ is zero as a pro-spectrum. Thus $\Delta_{\mathrm{TC}}(v;p) \simeq_{\mathrm{w}} 0$.

Finally put

$$K^{\mathrm{inv}} = \mathrm{fib}(K \longrightarrow \mathrm{TC}).$$

This is a truncating localizing invariant [13, Corollary 3.6, proof]. Every vertical map in the original formal four-vertex square is a π_0-isomorphism, so its relative K^{inv} term is zero. Naturality of the cyclotomic trace therefore identifies the vertical relative K-square with the vertical relative TC-square. After the same finite Zariski descent and total fiber,

$$\Delta_K(v;p) \simeq \Delta_{\mathrm{TC}}(v;p) \simeq_{\mathrm{w}} 0.$$

This proves both assertions. □

The truncation morphism $t_0X \to X$ is affine and, on an affine chart $\operatorname{Spec} A \subset X$, is induced by the 1-connective map $A \to H\pi_0 A$. Its base change preserves properness, the property of being locally almost of finite presentation, and the modification open. Moreover, because it is a π_0-isomorphism, strict base change carries each free Koszul stage to the corresponding free Koszul stage for the same classical center; hence the pulled-back tower is again cofinal. Taking $X' = t_0X$ in Theorem 10.4 therefore gives, for $E \in \{K, \mathrm{TC}\}$,

$$\Phi_E(p) \simeq_{\mathrm{w}} \Phi_E(Y \times_X t_0X \longrightarrow t_0X). \tag{10.7}$$

11 Proper spectral formal pro-excision

Theorem 11.1 (Proper spectral formal pro-excision)**.** *Let $p : Y \to X$ be a proper morphism, locally almost of finite presentation, between connective qcqs spectral schemes. Suppose that p is an equivalence over a quasi-compact open $U \subset X$, and put $Z = |X| \setminus U$. For $E \in \{K, \mathrm{TC}\}$ and the fixed cofinal free spectral Koszul towers, the square*

$$\begin{array}{ccc} E(X) & \longrightarrow & E(X^\wedge_Z) \\ \big\downarrow & & \big\downarrow \\ E(Y) & \longrightarrow & E(Y^\wedge_{p^{-1}Z}) \end{array} \tag{11.1}$$

is weakly cartesian in $\mathrm{Pro}(\mathrm{Sp})$*, where $E(X)$ and $E(Y)$ are regarded as constant pro-spectra. Equivalently, the canonical compression maps* (7.3) *form a weak equivalence of pro-spectra.*

Proof. Proposition 7.3 identifies formal pro-excision with the vanishing of the formal defect $\Phi_E(p)$. Proposition 8.4 proves this vanishing over a classical base, while Theorem 10.4 shows that the defect is unchanged by a 1-connective affine base change. We apply these two results in succession.

Set

$$X_0 = t_0X, \qquad Y_0 = Y \times_X X_0, \qquad p_0 : Y_0 \longrightarrow X_0.$$

The truncation morphism $X_0 \to X$ is affine; on every affine chart $\operatorname{Spec} A \subset X$, it is induced by the 1-connective map $A \to H\pi_0 A$. Choose the formal tower on X_0 by strict base change from the tower on X, and do the same on Y_0. Theorem 10.4 gives

$$\Phi_E(p) \simeq_{\mathrm{w}} \Phi_E(p_0), \qquad E \in \{K, \mathrm{TC}\}.$$

The morphism p_0 is proper and locally almost of finite presentation and is an equivalence over $U \cap X_0$. Its base is classical, so Proposition 8.4 makes $\Phi_E(p_0)$ weakly zero. Hence $\Phi_E(p)$ is weakly zero, which proves (11.1) and the equivalent compression statement. □

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